

Splunk-SOC-Lab

In this Practical SIEM & Detection Engineering episode, you'll deploy Splunk UF, configure receiving port 9997 on Splunk Enterprise, integrate Splunk TA for Windows, Atomic Red Team(ART) and Sysmon, and forward Windows Event Logs into the wineventlog index.

This hands-on lab is designed for SOC Analysts, Blue Teamers, and Cybersecurity learners building a professional Home SOC Lab.

In This You Will Learn:

- ✓ Install Splunk Universal Forwarder (UF) on Windows 10
- ✓ Configure Receiving Port 9997 in Splunk Enterprise
- ✓ Create and use index wineventlog
- ✓ Install Splunk TA for Windows
- ✓ Install Splunk TA for Sysmon
- ✓ Deploy Sysmon using SwiftOnSecurity configuration
- ✓ Configure inputs.conf and outputs.conf
- ✓ Allow firewall ports (Host & VM)
- ✓ Set UF service to Local System Account
- ✓ Atomic Red Team(ART)
- ✓ Validate log ingestion and build basic visualizations

This lab is essential for building a real SIEM environment and improving Detection Engineering skills.

Perfect for SOC Analyst preparation, Blue Team practice, and Cybersecurity Home Labs.

Keyword Block

- Splunk
- Splunk Universal Forwarder
- Splunk UF Windows 10
- Splunk Sysmon integration
- SIEM Home Lab
- SOC Home Lab setup
- Splunk receiving port 9997
- Splunk inputs.conf configuration
- Splunk outputs.conf configuration
- Splunk TA Windows
- Atomic Red Team(ART)
- Splunk TA Sysmon
- Sysmon SwiftOnSecurity
- Windows Event Log monitoring
- wineventlog index
- Detection Engineering lab
- Blue Team lab

- Cybersecurity SIEM tutorial
- Splunk installation guide
- Install Splunk UF
- Splunk Enterprise configuration
- Log analysis with Splunk
- Home SOC lab
- Practical SIEM tutorial.

Lab Goals

The objective of this lab is to:

- Build a functional SIEM environment
- Collect and analyze Windows Event Logs
- Forward logs using Splunk Universal Forwarder
- Deploy Sysmon for enhanced telemetry
- Simulate attacks using Atomic Red Team
- Detect suspicious activity inside Splunk Enterprise
- Practice Detection Engineering and Blue Team monitoring

Environment Architecture

Design the lab using:

- Host Machine
- Windows 10 VM
- Kali Linux attacker VM
- Splunk Server
- Windows Endpoint with Sysmon + UF installed

Include:

- Network diagram
- IP addressing example
- NAT vs Bridged networking explanation
- Hardware requirements
- Recommended VM settings

Technologies Used

Explain and configure:

- Splunk Enterprise
- Splunk Universal Forwarder
- Sysmon
- SwiftOnSecurity
- Splunk Add-on for Microsoft Windows
- Splunk Add-on for Sysmon
- Atomic Red Team
- MITRE ATT&CK Framework
- Windows Event Logging
- Detection Engineering concepts

How To Install Splunk Enterprise on Linux

What is Splunk?

[Splunk Official Website](https://splunk.com)

Splunk is a data analytics and Security Information and Event Management (SIEM) platform that collects, indexes, searches, analyzes, and visualizes machine-generated data from servers, applications, networks, cloud services, and security tools.

It is widely used for:

- Security monitoring
- Threat detection
- Incident response
- Log management
- IT operations monitoring
- Compliance reporting
- Threat hunting

Why Learn Splunk?

For cybersecurity professionals, SOC analysts, IT auditors, and security engineers, Splunk is one of the most valuable skills because it enables:

- Centralized log collection
- Real-time monitoring
- Alerting and correlation
- Dashboard creation
- Threat hunting
- MITRE ATT&CK mapping
- Security investigations

As a CISA-certified professional moving toward IT Audit and Cybersecurity, learning Splunk can significantly strengthen your technical security assessment capabilities.

Splunk on Kali Linux

[Splunk Enterprise Downloads](#)

Splunk Enterprise can be installed on Kali Linux because Kali is based on Linux. Many cybersecurity students and professionals use Kali as a lab environment for:

- Splunk Enterprise
- Sysmon log analysis
- Atomic Red Team testing
- Wazuh integration
- Detection engineering
- Threat hunting

A common home lab architecture is:

Windows Endpoint
├── Sysmon
├── Splunk Universal Forwarder
└── Atomic Red Team

↓

Kali Linux
└─ Splunk Enterprise

↓

Dashboards, Alerts & Threat Hunting

Benefits of Splunk 10.x

Recent Splunk Enterprise releases have focused on:

- Improved search performance
- Better resource utilization
- Enhanced cloud integration
- Expanded data management capabilities
- Improved security monitoring features
- Better support for hybrid and cloud environments

Always review the official release notes for the exact features of a specific version because capabilities can change between releases.

Recommended Cybersecurity Learning Path

Since you are already working toward cybersecurity and threat detection skills:

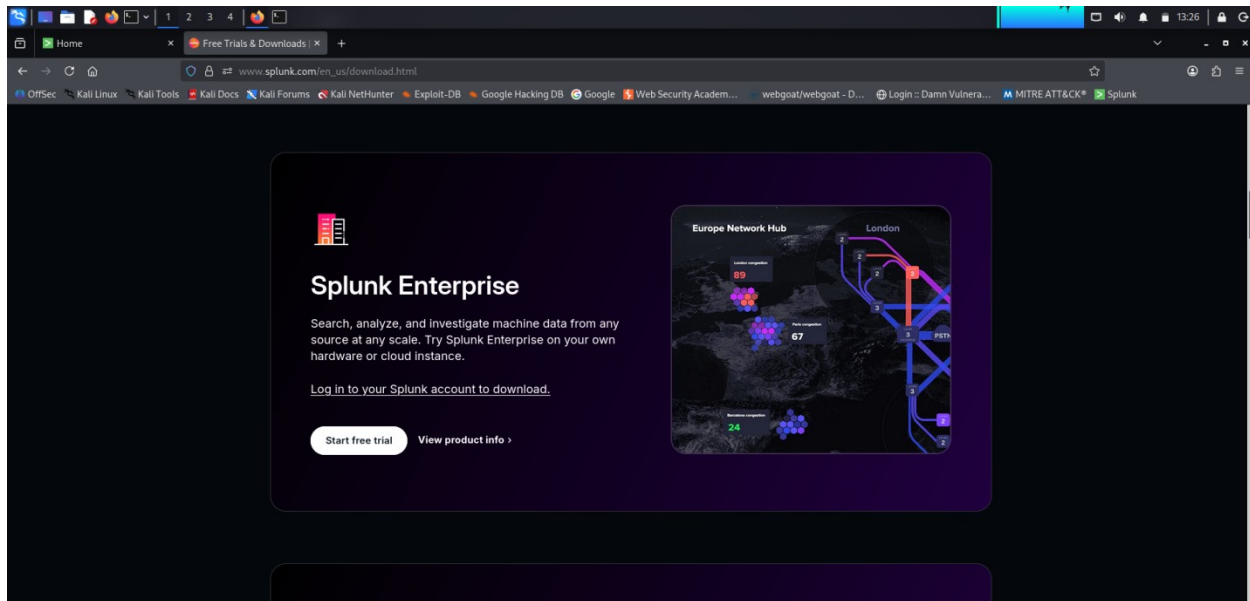
1. Install Splunk Enterprise on Kali Linux.
2. Install Sysmon on a Windows VM.
3. Install Splunk Universal Forwarder.
4. Forward Sysmon logs to Splunk.
5. Install Atomic Red Team.
6. Run MITRE ATT&CK simulations.
7. Create Splunk detections and dashboards.
8. Add Wazuh for endpoint detection and response.

This lab setup closely resembles what many SOC analysts, threat hunters, and detection engineers use for practical training.

In this guide, I'll walk you through the complete process of installing Splunk on Kali Linux using the official Debian package.

What You Need Before Starting

- A Kali Linux system (bare metal or VM)
- Internet connection
- Splunk Enterprise .deb file downloaded from the official Splunk website

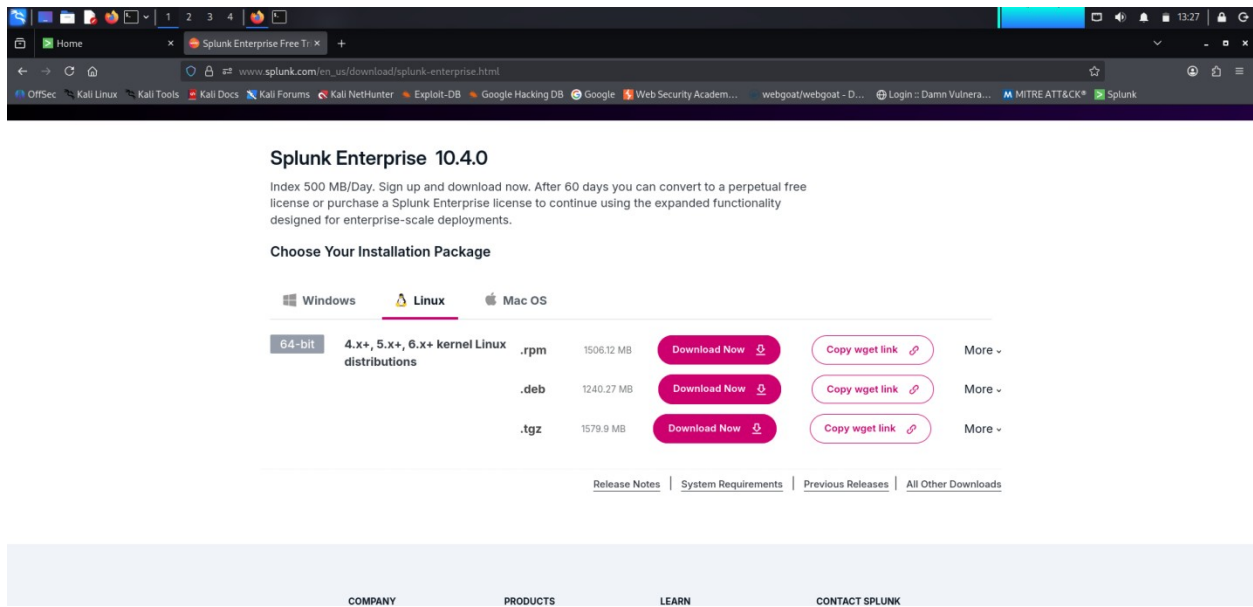


Step 1: Download Splunk Enterprise

Step 1: Go to the official website of Splunk or click on the link below to redirect to the website and follow the procedure below.

[Splunk Enterprise Download Page](#) Download the **Linux .deb** package.

Example file: splunk-10.4.0-linux-amd64.deb



Installation of Splunk for the First Time

Step 1 : Locate the File

To install Splunk, go to the folder where you downloaded the file. Usually, the file is in the Downloads folder. Open the terminal and change to the Downloads folder. Use the below command.

Step 2: Install the Splunk Debian Package

Command :

cd Downloads

```
(abhin@kali)~[~]
$ cd Downloads

(abhin@kali)~[/Downloads]
$ ls
'100 Web Vulnerabilities List.pdf'
'662631325003dd9fefdbae6dcb4807fc6d856baf'
'662631325003dd9fefdbae6dcb4807fc6d856baf.zip'
'abhinsubej(1).ovpn'
'Abhinsubej VAPT Project-1.pdf'
'Blue Team Junior Analyst Pathway Bundle-btja.pdf'
cacert.der
completion_certificate.pdf
Configurations.docx
converted_text-1.pdf
create.sql
'CSA ILABS user guide ... (.pdf'
CSA_MERGED.pdf
Frame_1161_a1717e1c65.jpg
introduction_to_cip.png
'Introduction to Dark Web Operations-course.pdf'
'Introduction to Digital Forensics-course.pdf'
'Introduction to OSINT-course.pdf'
'Introduction to Vulnerability Management-course.pdf'
kome-text.pdf

(abhin@kali)~[/Downloads]
$ sudo dpkg -i splunk-10.0.2-e2d18b4767e9-linux-amd64.deb_
```

Step 3 : Change the Directory

After extracting the Splunk installation files, we need to change to the specific directory where the program files are located. We can do this by using the following command.

Command :

```
cd /opt/splunk/bin
```

After installation, go to the Splunk bin folder:

Command: cd /opt/splunk/bin

This folder contains all Splunk executable files such as:

- splunk start
- splunk stop
- splunk restart
- splunk status

```
(kali@kali)-[~/Downloads]
$ cd /opt/splunk/bin
```

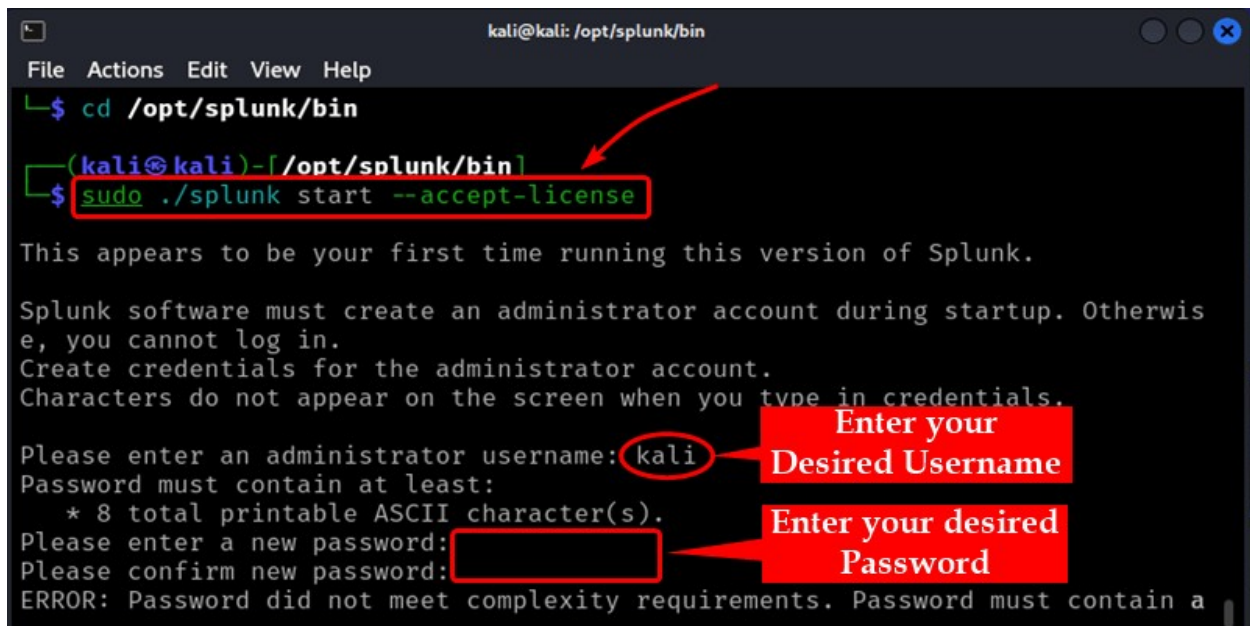
```
(root@kali)-[/opt/splunk/bin]
# ls
2to3-3.7          genSignedServerCert.py  node                python              slim
bloom             genSignedServerCert.sh  openssl             python3             spl2-orchestrator
bottle.py         genWebCert.py           parse_xml_buckets.py python3.7            spl-lang-server-sockets
btool            genWebCert.sh           pcre2-config        pyvenv              splunk
btprobe          idle3                   pcregextest         pyvenv-3.7          splunkd
bzip2            idle3.7                 pid_check.sh        rapiddiag            splunkmon
classify         importtool              pip3                 recover-metadata    splunk-optimize
ColdStorageArchiver_GCP.py installit.py            pip3.7              rest_handler.py     splunk-optimize-lex
ColdStorageArchiver.py jars                    prichunkpng         runScript.py        splunk-tlsd
coldToFrozenExample.py jsmin                  priforgepng         S3benchmark         supervisor-simulator
compsup          locktest                pripgreypng         safe_restart_cluster_master.py tarit.py
copyright.txt    locktool               pripalpng            scripts              tocsv.py
dbmanipulator.py mongod                  pripnglsch          scrubber.py          tsidxprobe
easy_install-3.7 mongod-3.6             pripngtopam         searchtest           tsidxprobe_plo
exporttool       mongod-4.0             priweavepng         setSplunkEnv         tsidx_scan.py
fill_summary_index.py mongorestore           pydoc3              shc_upgrade_template.py untarit.py
genAuditKeys.py  noah_self_storage_archiver.py pydoc3.7            signtool             walklex
genRootCA.sh
```

Step 4 : Accept the License & Create Username and Password

After changing to the Splunk program directory, we need to accept the license agreement to start the Splunk server. We can do this by running the following command. After running this command, Splunk will prompt you to enter a username and password. You can choose any username and password you prefer. This username and password will be used to log into the Splunk web interface later.

Command :

```
sudo ./splunk start --accept-license
```

A terminal window titled 'kali@kali: /opt/splunk/bin' showing the execution of 'sudo ./splunk start --accept-license'. The command is highlighted with a red box and an arrow. The output shows the Splunk startup process, including prompts for an administrator username (with 'kali' entered and circled) and a password. Two red callout boxes with white text provide instructions: 'Enter your Desired Username' pointing to the username input, and 'Enter your desired Password' pointing to the password input field. The terminal also shows a message about password complexity requirements.

```
kali@kali: /opt/splunk/bin
File Actions Edit View Help
└─$ cd /opt/splunk/bin
└─(kali@kali)-[/opt/splunk/bin]
└─$ sudo ./splunk start --accept-license

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: kali
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
ERROR: Password did not meet complexity requirements. Password must contain a
```

Step 5 : Access Splunk server in Browser

After setting up the username and password, you will see a web address displayed in the terminal output. This is the address you need to access the Splunk server through a web browser.

```
kali@kali: /opt/splunk/bin
File Actions Edit View Help
PYTHONHTTPSVERIFY is set to 0 in splunk-launch.conf disabling certificate validation for the httplib and urllib libraries shipped with the embedded Python interpreter; must be set to "1" for increased security
Done

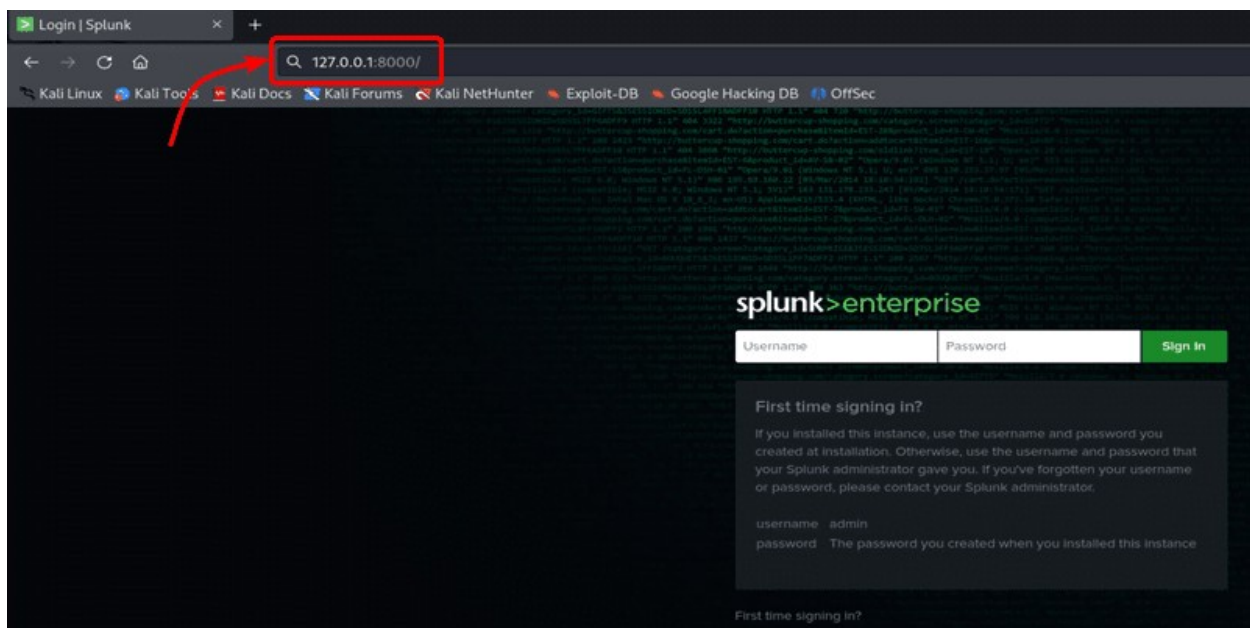
Waiting for web server at http://127.0.0.1:8000 to be available.....
.. Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://kali:8000

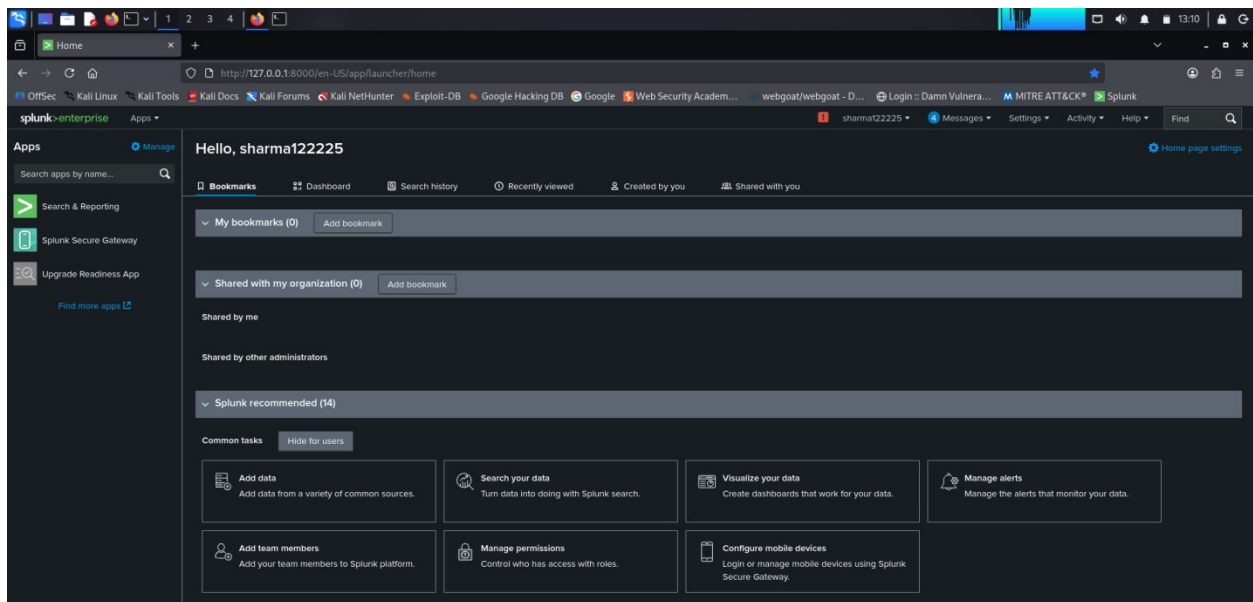
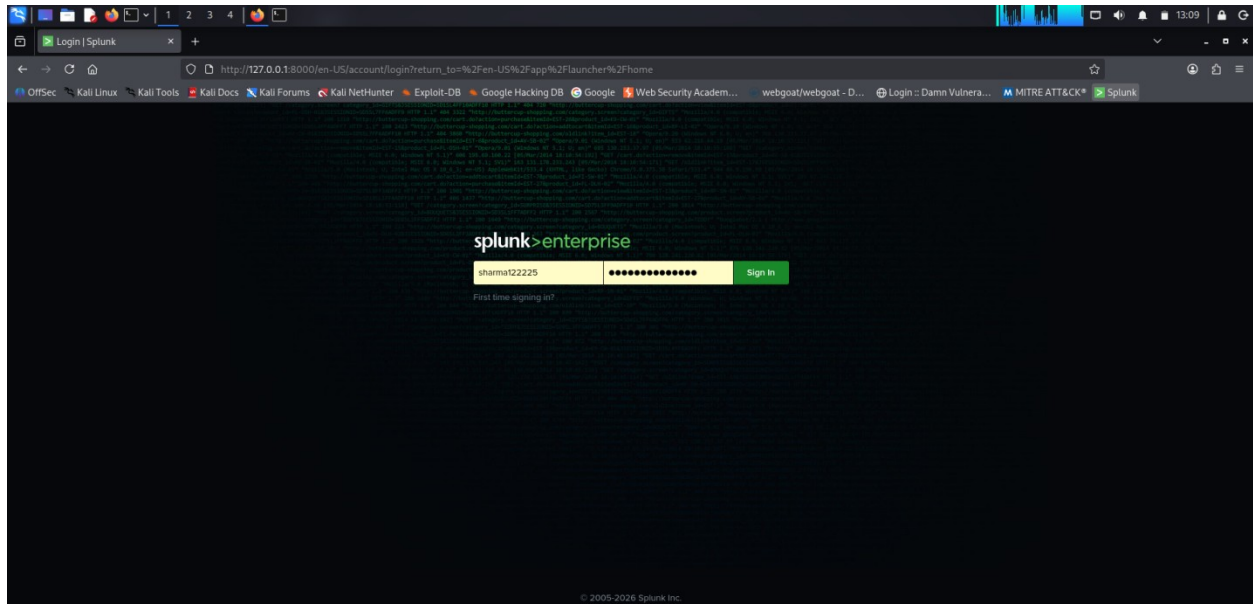
(kali@kali)-[/opt/splunk/bin]
$
```

Enter this Web Address in your Browser to access the Splunk Server



Step 6 : Sign in to the Splunk Server

On the Splunk login page, you will see fields to enter your username and password. Enter the same username and password that you set in Step 4 when you ran the command to start the Splunk server.



That's it! You have now successfully signed in to the Splunk server using the web interface. After signing in, you will be able to access the Splunk dashboard and start exploring its various features for searching, analyzing, and visualizing your data.

Step 7. Configure Receiving Port 9997

- Enable receiving- Settings → Forwarding and Receiving
- Configure forwarder management
- Verify listening port-9997
- Save.

- Verify Port

Run: `sudo ss -tulnp | grep 9997`
9997 LISTEN

Step 8. Create Splunk Index

Go to:
Settings → Indexes

Create new index:

Wineventlog

The screenshot shows the Splunk Enterprise web interface. A dialog box titled 'Edit Index: wineventlog' is open, displaying various settings for the 'wineventlog' index. The settings include Home Path, Cold Path, Thawed Path, Data Integrity Check, Max Size of Entire Index, Max Size of Hot/Warm/Cold Buckets, and Frozen Path. The 'Indexes' list on the right shows the 'wineventlog' index with a size of 500 GB and a status of 'Enabled'.

Edit Index: wineventlog

General Settings

Home Path:
Hot/warm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/db)

Cold Path:
Cold db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/colddb)

Thawed Path:
Thawed/unconnected db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/thaweddb)

Data Integrity Check: ☐ Enable ☒ Disable
Enable this if you want Splunk to compute hashes on every slice of your data for the purpose of data integrity.

Max Size of Entire Index: 500 GB
Maximum target size of entire index.

Max Size of Hot/Warm/Cold Buckets: auto MB
Maximum target size of buckets. Enter 'auto_high_volume' for high-volume indexes.

Frozen Path:
Frozen bucket archive path. Set this if you want Splunk to automatically archive frozen buckets.

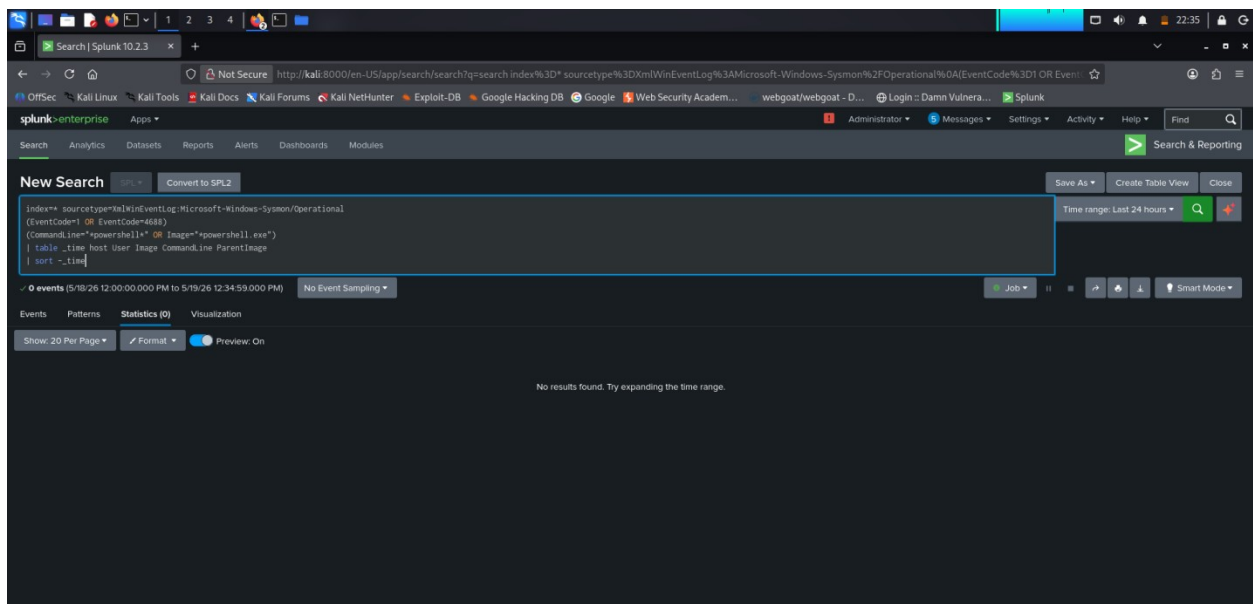
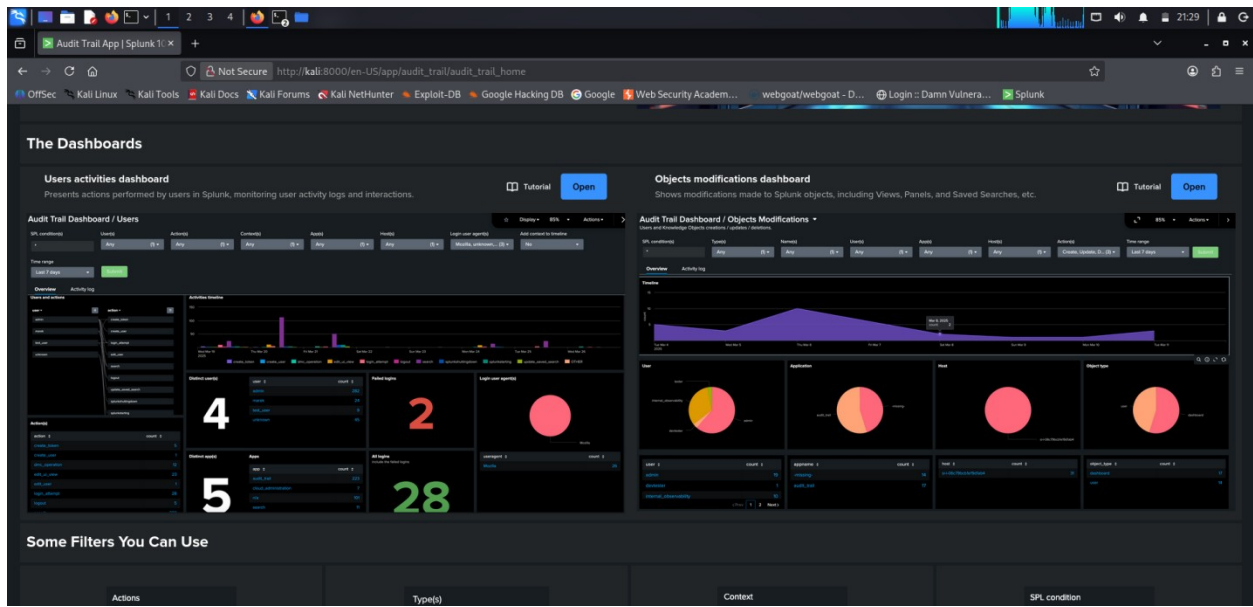
App:

Indexes

A repository for data in Splunk Enterprise. Indexes reside in flat files on the Splunk Enterprise instance known as the index. [Learn more](#)

16 Indexes

Name	Actions	Type	App	Current Size	Max Size	Event Count	Earliest Event	Latest Event	Home Path	Frozen Path	Status
._audit	Edit Delete Disable	Events	system	2 MB	488.28 GB	10.4K	15 days ago	14 days ago	\$SPLUNK_DB/_audit/db	N/A	✓ Enabled
._configtracker	Edit Delete Disable	Events	system	3 MB	488.28 GB	211	15 days ago	14 days ago	\$SPLUNK_DB/_configtracker/db	N/A	✓ Enabled
._disappevent	Edit Delete Disable	Events	SplunkDeploymentServerConfig	1 MB	488.28 GB	0			\$SPLUNK_DB/_disappevent/db	N/A	✓ Enabled
._dsclient	Edit Delete Disable	Events	SplunkDeploymentServerConfig	1 MB	488.28 GB	0			\$SPLUNK_DB/_dsclient/db	N/A	✓ Enabled
._dsphonehome	Edit Delete Disable	Events	SplunkDeploymentServerConfig	1 MB	488.28 GB	0			\$SPLUNK_DB/_dsphonehome/db	N/A	✓ Enabled
._internal	Edit Delete Disable	Events	system	2 MB	488.28 GB	13.1K	15 days ago	14 days ago	\$SPLUNK_DB/_internal/db	N/A	✓ Enabled
._introspection	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/_introspection/db	N/A	✓ Enabled
._metrics	Edit Delete Disable	Metrics	system	1 MB	488.28 GB	0			\$SPLUNK_DB/_metrics/db	N/A	✓ Enabled
._metrics_rollup	Edit Delete Disable	Metrics	system	1 MB	488.28 GB	0			\$SPLUNK_DB/_metrics_rollup/db	N/A	✓ Enabled
._telemetry	Edit Delete Disable	Events	system	1 MB	488.28 GB	1	14 days ago	14 days ago	\$SPLUNK_DB/_telemetry/db	N/A	✓ Enabled
._thefishbucket	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/_fishbucket/db	N/A	✓ Enabled
._history	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/_history/db	N/A	✓ Enabled
._main	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/_default/db	N/A	✓ Enabled
._splunklogger	Edit Delete Disable	Events	system	0 B	488.28 GB	0			\$SPLUNK_DB/_splunklogger/db	N/A	✗ Disabled
._summary	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/_summary/db	N/A	✓ Enabled
._wineventlog	Edit Delete Disable	Events	search	1 MB	500 GB	0			\$SPLUNK_DB/_wineventlog/db	N/A	✓ Enabled



Essential Management Commands

Start Splunk

```
sudo /opt/splunk/bin/splunk start
```

Stop Splunk

```
sudo /opt/splunk/bin/splunk stop
```

Restart Splunk

```
sudo /opt/splunk/bin/splunk restart
```

Check Status

```
sudo /opt/splunk/bin/splunk status
```

Verification Checklist

Task	Status
Download .deb file	✓
Install Splunk Enterprise	✓
Accept License	✓
Create Admin User	✓
Enable Boot Start	✓
Open Web Interface	✓
Login Successfully	✓

Final Thoughts

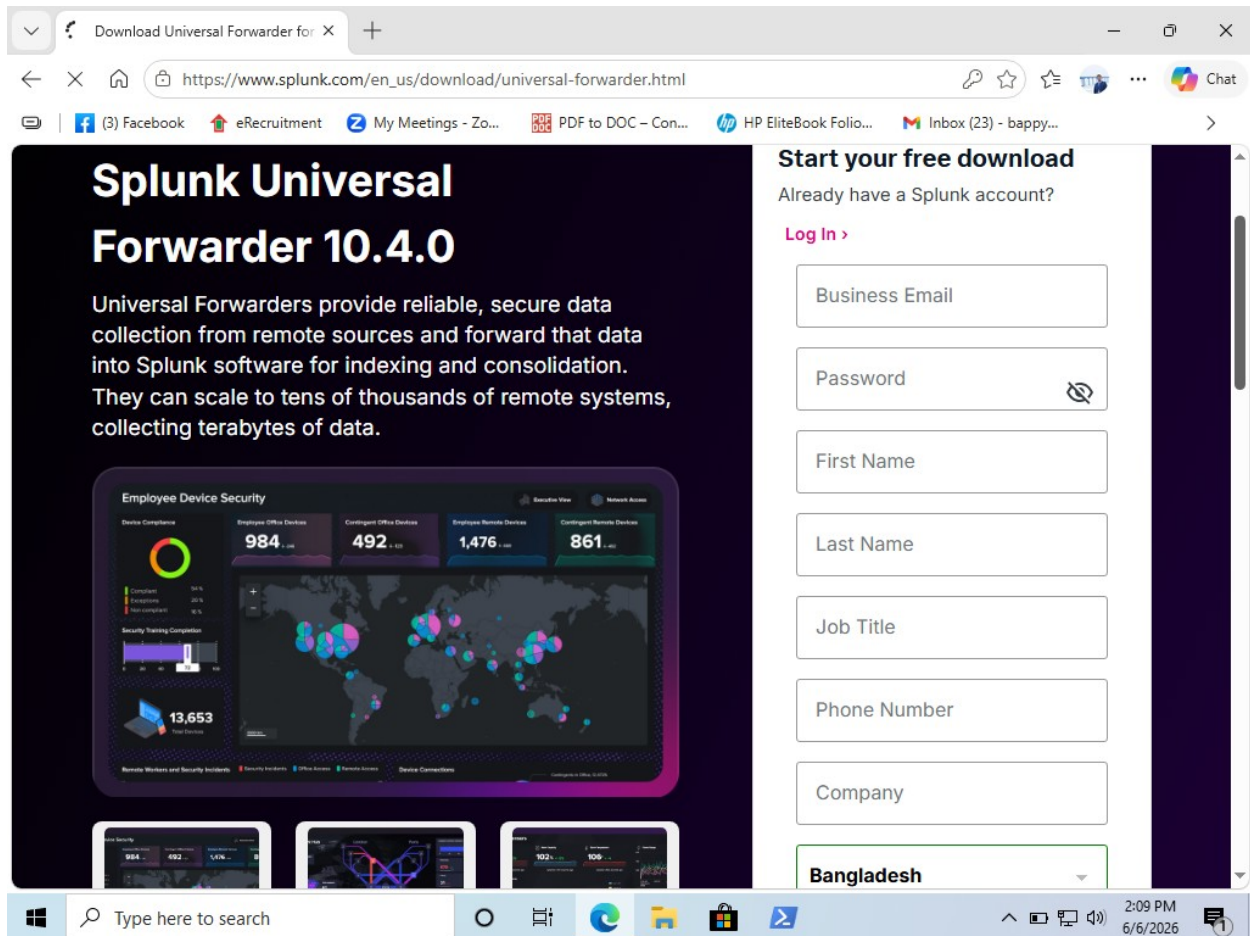
- Setting up Splunk on Kali Linux allows you to:
- Practice SIEM concepts
- Analyze logs
- Learn detection engineering
- Prepare for SOC analyst roles
- Build threat-hunting skills

This installation method is clean, simple, and works across most Debian-based systems.

Conclusion

Getting Splunk up and running on your Linux machine is not too difficult. After downloading the Splunk file, you extract it and use some basic terminal commands to start the Splunk server. Splunk will then give you a web address to access its interface through your browser. Just create a username and password when asked, open that web address, and sign in with your new credentials. With those straightforward steps completed, you now have Splunk installed and ready to use for collecting and analyzing data on your Linux system.

Install Splunk Universal Forwarder on Windows Victim Machine



Installation Steps

Obtain the Installation Package

First, download the Splunk Universal Forwarder from Splunk's download page. You will need a Splunk.com account to access the download. In the event you need to download an older version of the Universal Forwarder, those packages are available on the older releases page.

For this process, you'll want to download the MSI package for your version of Windows.

GET STARTED

Choose Your Download

Splunk Universal Forwarder 8.1.2

Universal Forwarders provide reliable, secure data collection from remote sources and forward that data into Splunk software for indexing and consolidation. They can scale to tens of thousands of remote systems, collecting terabytes of data.

Choose Your Installation Package

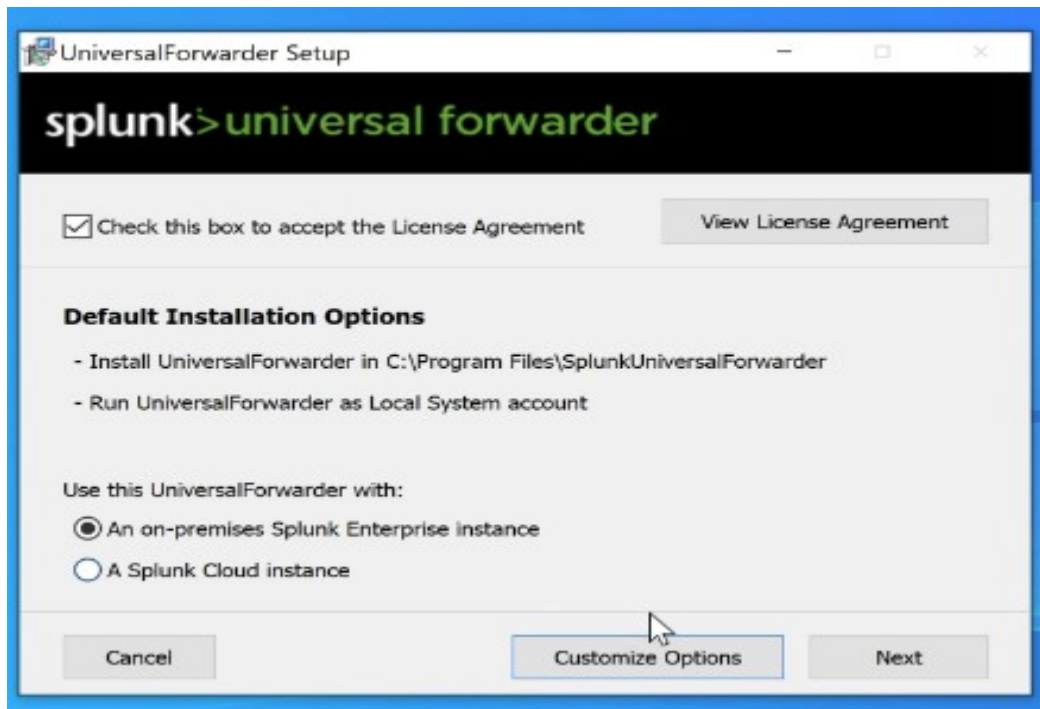
 Windows	 Linux	 Solaris	 Mac OS	 FreeBSD	 AIX
64-bit	Windows 10 Windows Server 2016, 2019			.msi 70.5 MB	Download Now 
32-bit	Windows 10			.msi 59.04 MB	Download Now 

[Release Notes](#) | [Older Releases](#) | [All Other Downloads](#)

Using the GUI Installer

Double-click the MSI file to start the installation.

The first screen of the installer should pop up. Select the Check this box to accept the License Agreement check box and the check box for A Splunk Cloud instance.

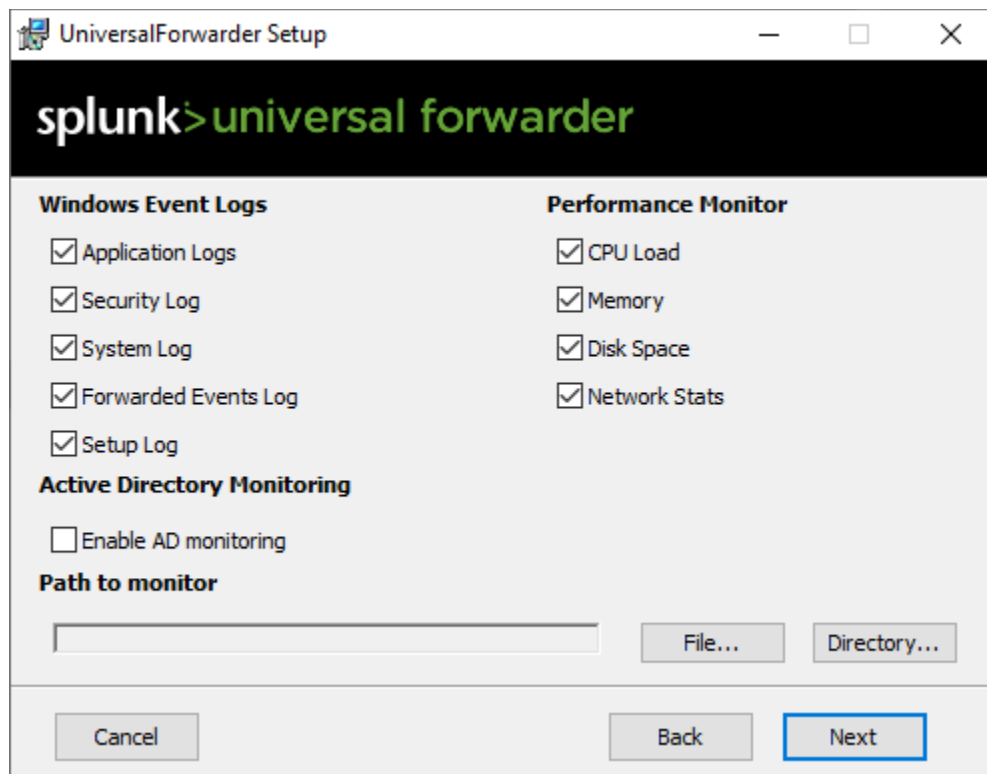


Select Data to Forward

Under these sections:

- Windows Event Logs
- Performance Monitor

Select all options. Click Next.



This determines what logs will be sent to your SOC.

Credentials

Set a username and password. Click Next.

- As I have locally deployed SPLUNK on same machine of which I'm forwarding logs, so that host name will be as follows in my case and by default port is 8089

Note: You will be using the IP of the machine on which you have installed the Splunk and you can get the IP with the "ipconfig" command and use that in the installation process of universal forwarder

splunk>universal forwarder

If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

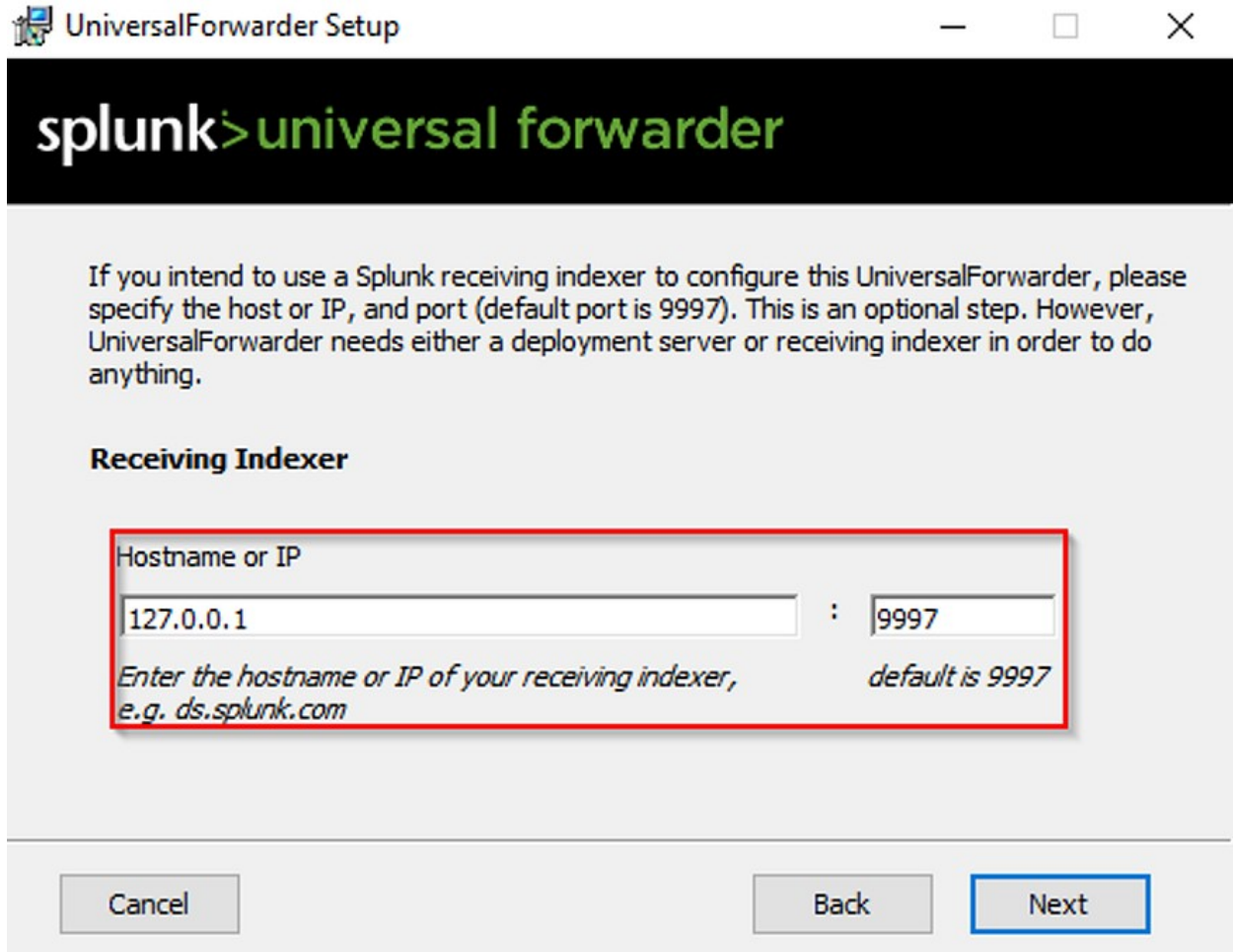
Deployment Server

Hostname or IP

:

*Enter the hostname or IP of your deployment server,
e.g. ds.splunk.com*

default is 8089



Configure Receiving Indexer
This is the important part.

Enter:

- The IP address of the SOC machine
- Port:

9997

This tells the forwarder exactly where to send logs.

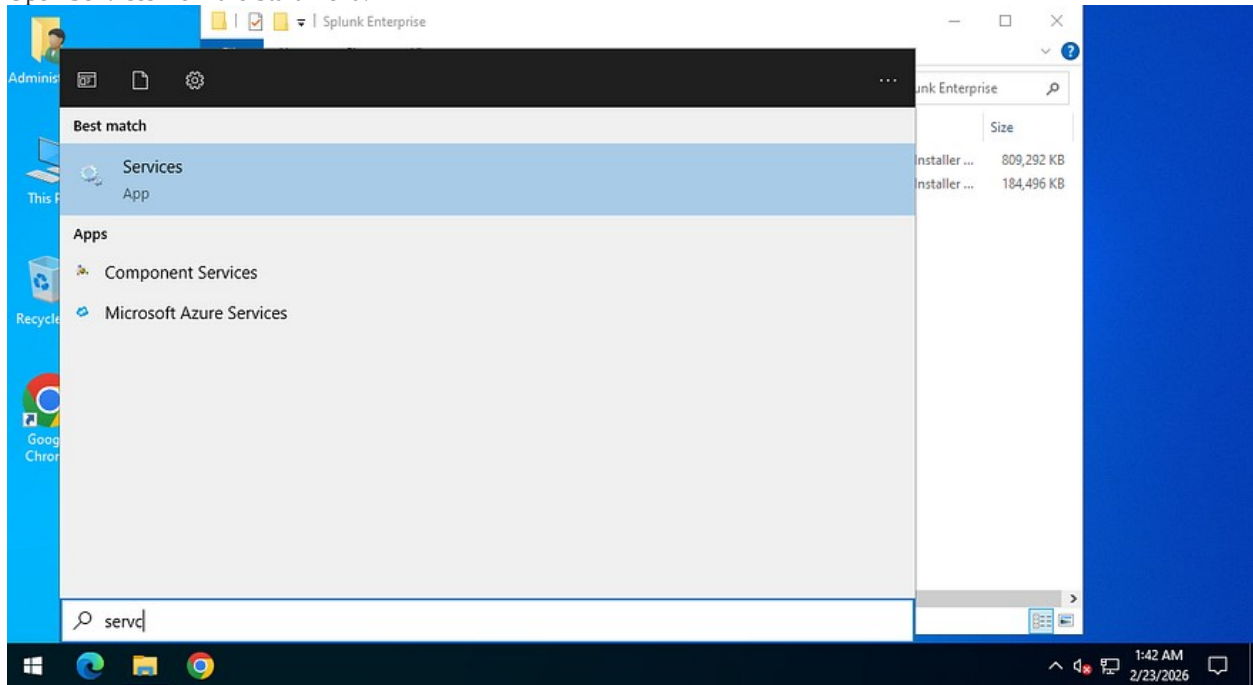
Click Next, then Install.

After installation completes, click Finish.

Restart the Forwarder Service

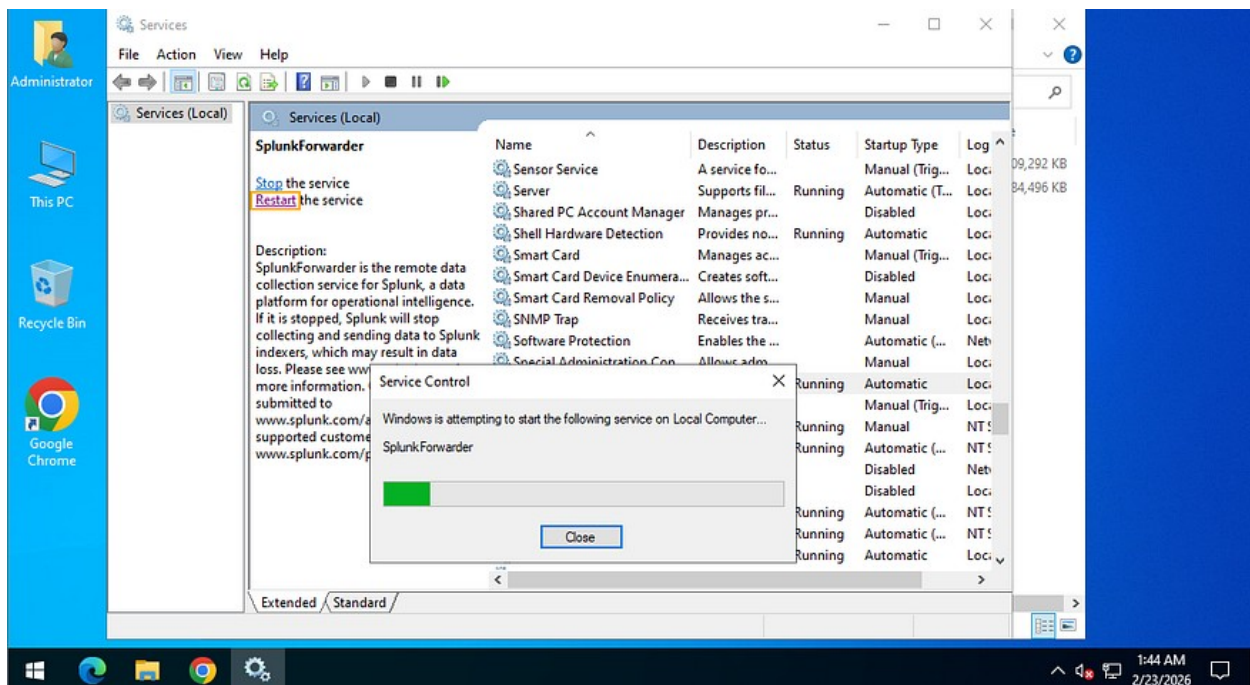
On the Windows Server:

Open **Services** from the Start menu.



Scroll down to SplunkForwarder.

Click it and select Restart.



Verify the service is running:

sc query SplunkForwarder

You should see STATE: 4 RUNNING.

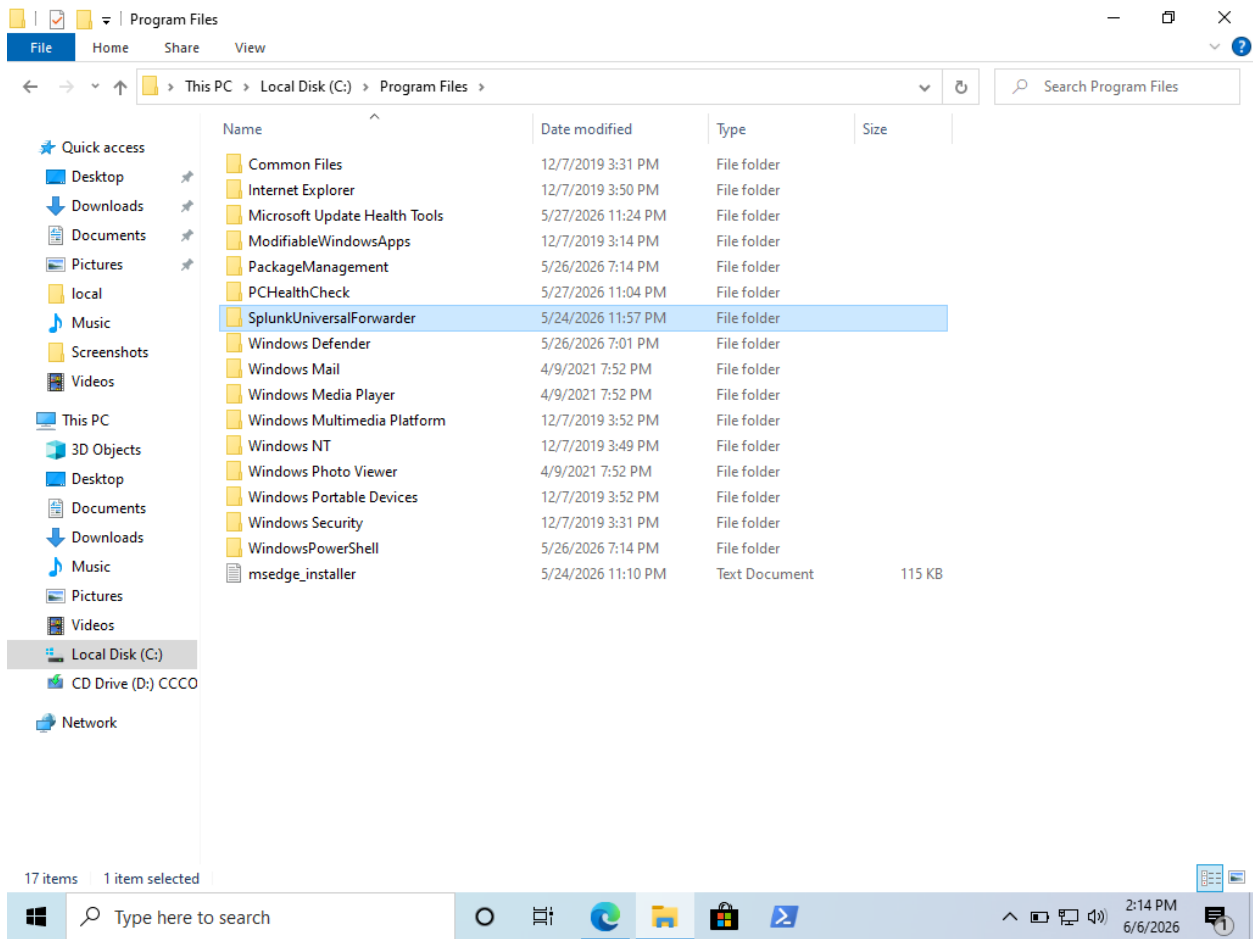
Smart Card Removal Policy	Allows the s...	Manual	Local Syste...
SNMP Trap	Receives tra...	Manual	Local Service
Software Protection	Enables the ...	Automatic (D...	Network S...
SplunkForwarder	SplunkForw...	Running Automatic	NT SERVIC...
Spot Verifier	Verifies pote...	Manual (Trig...	Local Syste...

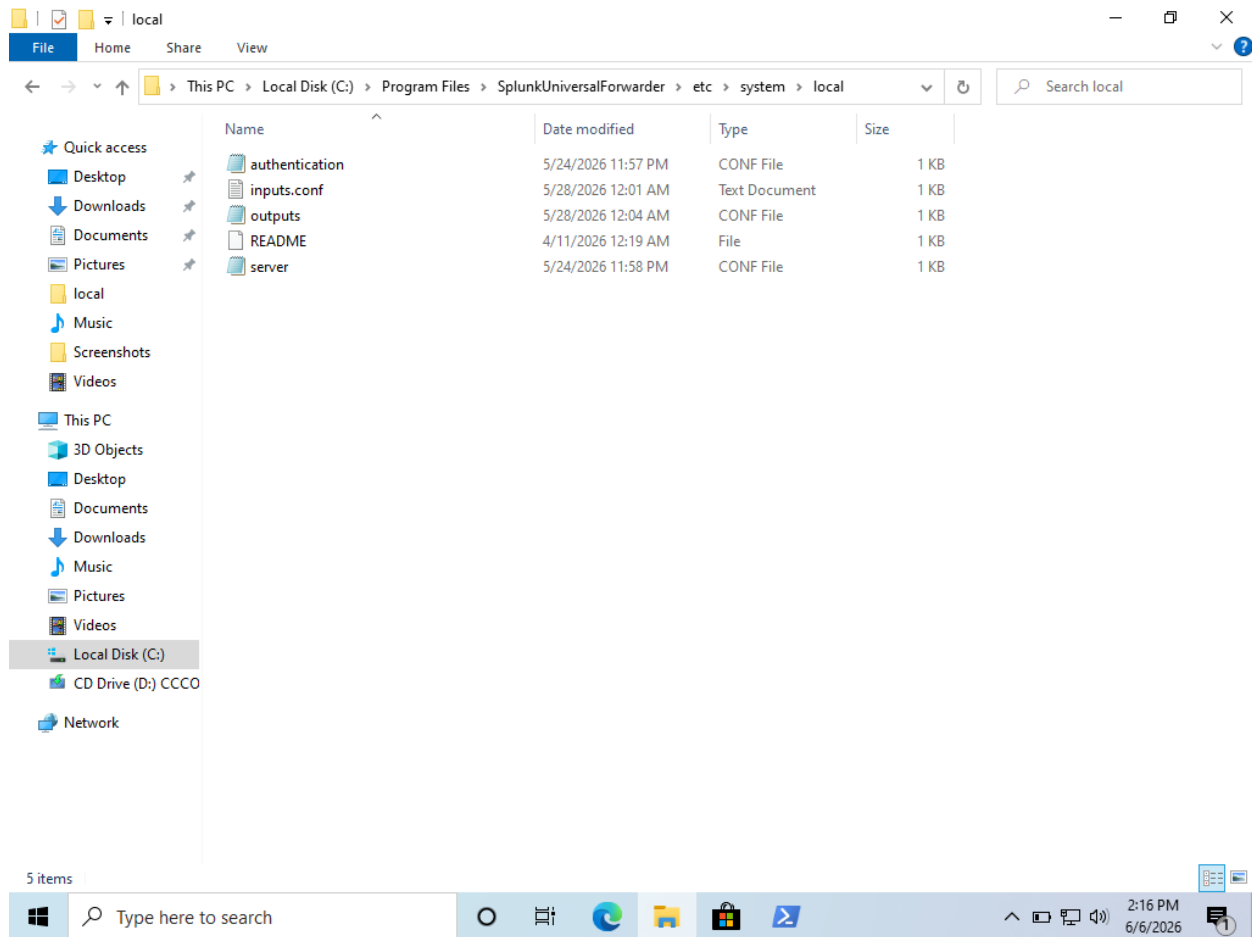
Step 3: Configure inputs.conf for Log Collection

Now we need to tell the forwarder which logs to collect. This is done through the inputs.conf file.

Navigate to the configuration directory:

cd "C:\Program Files\SplunkUniversalForwarder\etc\system\local"





Configure outputs.conf

Open Path:

C:\Program Files\SplunkUniversalForwarder\etc\system\local\

Provide professional examples such as:

[tcpout]

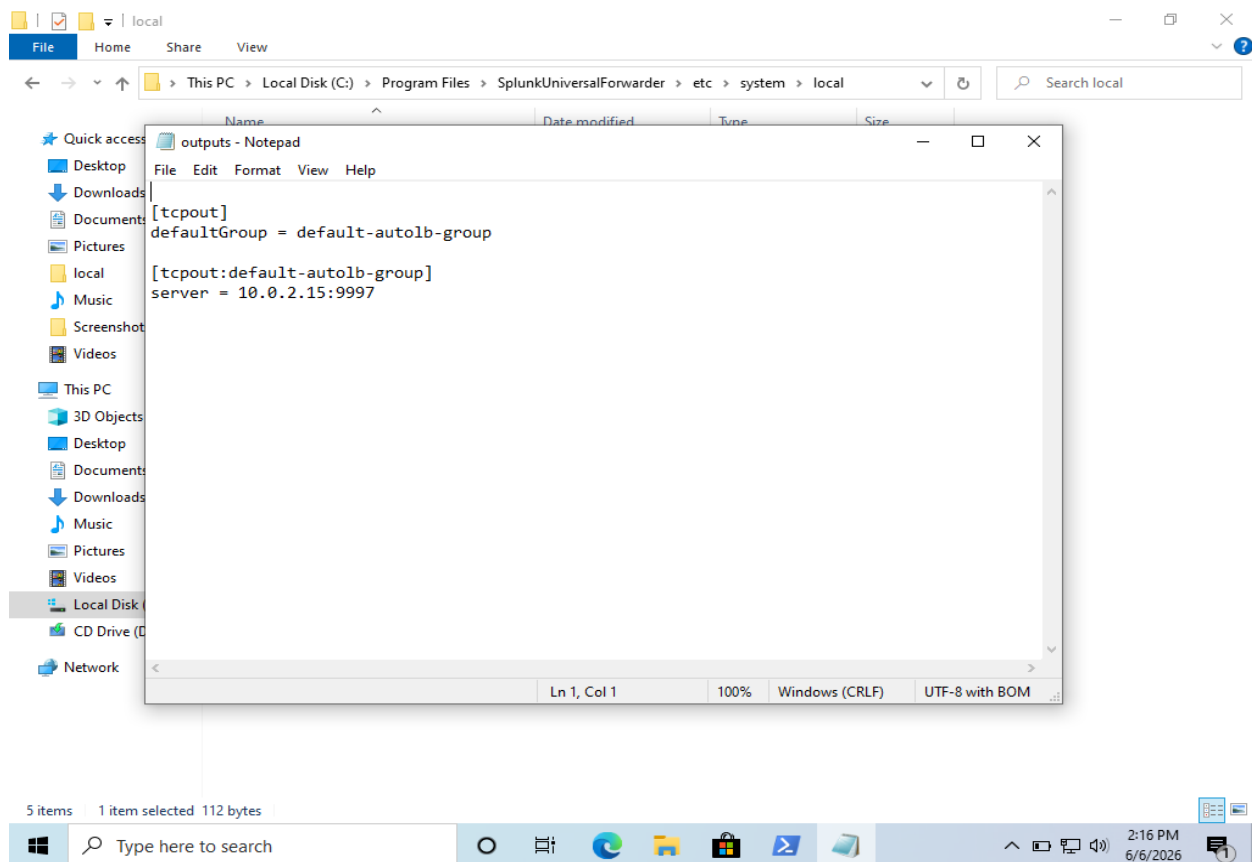
defaultGroup = default-autolb-group

[tcpout: default-autolb-group]

server = ip:9997

[tcpout-server: //ip:9997]

Explain each configuration line.



Configure inputs.conf

Create:

inputs.conf

Path:

C:\Program Files\SplunkUniversalForwarder\etc\system\local\

Add:

[WinEventLog:// Application]

disabled = 0

index = wineventlog

[WinEventLog:// Security]

disabled = 0

index = wineventlog

[WinEventLog:// System]

disabled = 0

index = wineventlog

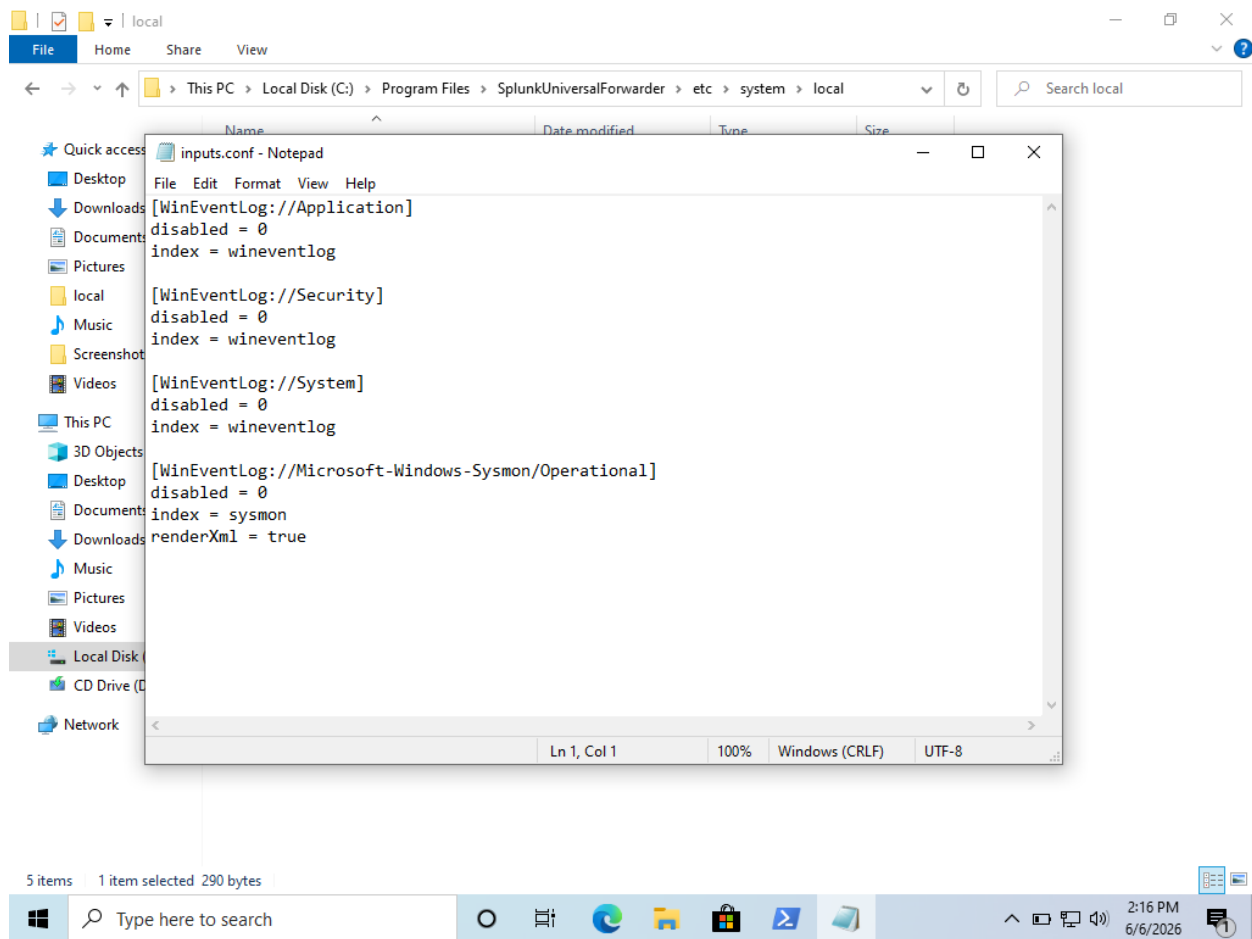
[WinEventLog://Microsoft-Windows-Sysmon/Operational]

disabled = 0

index = wineventlog

Restart:

Splunk list forward-server.



Install Splunk TA for Windows

Explain:

- Purpose
- Field extraction
- CIM compatibility

Install Splunk TA for Sysmon

Configure:

Download:

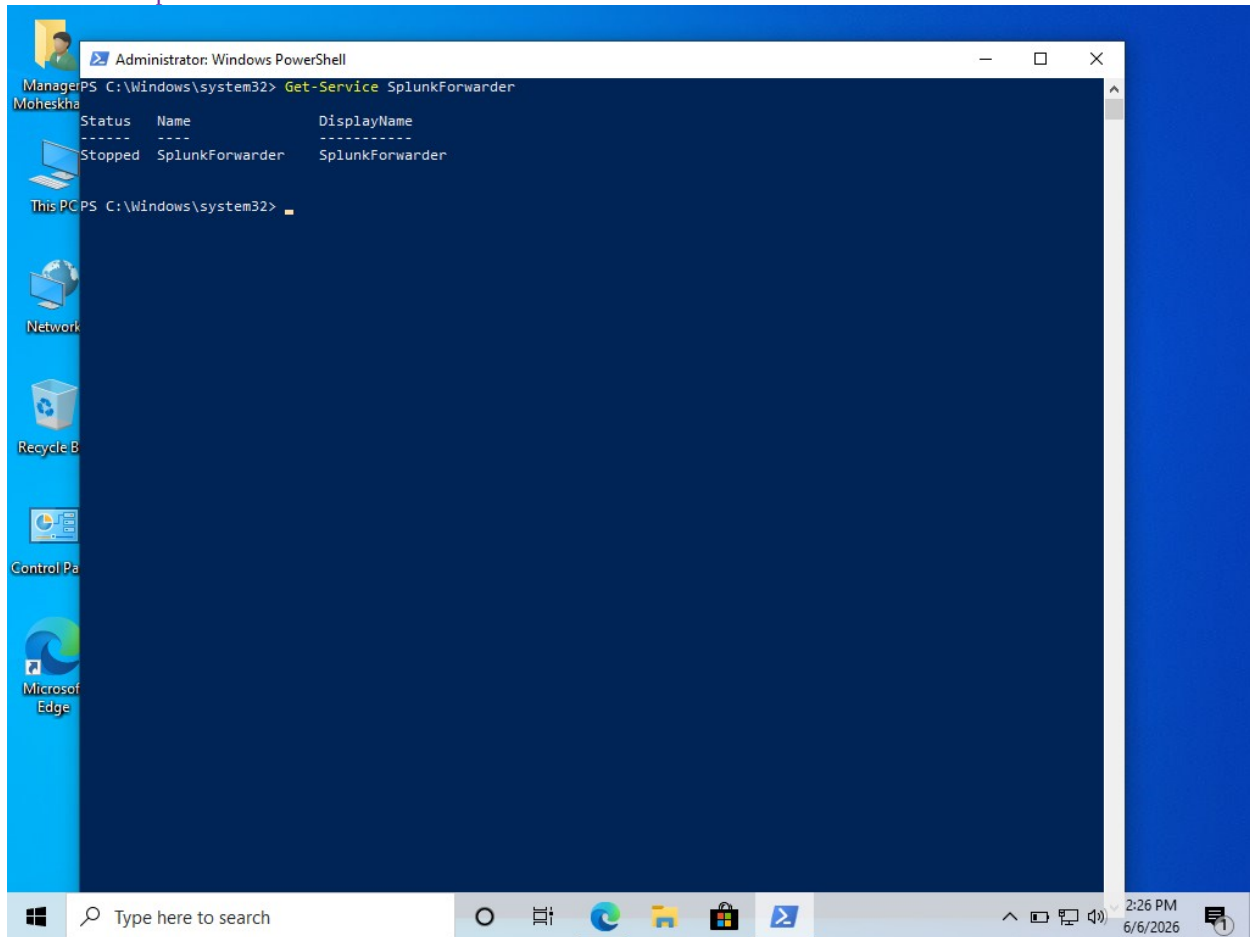
Splunk Add-on for Sysmon

Install into:

/opt/splunk/etc/apps/

Restart-service SplunkForwarder

Get-Service SplunkForwarder



Troubleshooting Guide

Explain solutions for:

- Inactive forwards- Command: `sc query splunkforwarder`
- No events in Splunk- Command: `netstat -ano | findstr 9997`
- Port connectivity problems
- Incorrect sourcetypes
- Sysmon not generating logs
- UF service failures
- Firewall blocks
- Splunk Server:
Command:
`sudo /opt/splunk/bin/splunk add forward-server ip:9997`
`sudo /opt/splunk/bin/splunk list forward-server`

```
sudo netstat -tulnp | grep 9997
```

- Restart and others command:
splunk start
splunk stop
splunk restart
splunk status
splunk list forward-server
splunk add forward-server ip*:9997
- Check Splunk Logs- Command: `sudo tail -f /opt/splunk/var/log/splunk/splunkd.log`

What this does:

- Sets the hostname for all events
- Collects Windows Security, System, and Application logs → windows_security index
- Collects Sysmon logs → sysmon index
- renderXml = true ensures Sysmon data is properly parsed
- Save the file and close Notepad.
- *It will take around 2 to 3 minutes to finish the installation.*

And That's the Foundation

Getting Started with Sysmon: Installation and Configuration

What is Sysmon

Sysmon makes it possible to monitor activities on the Windows operating system in detail. It provides detailed information on the created network connections, file changes, registry activities, or created processes. Sysmon can be used in combination with Defender for Endpoint. Microsoft is not logging all events with Defender for Endpoint – with the addition of Sysmon, it is possible to ingest more events and add additional events for investigation.

Sysmon is free and part of the Sysinternals software package. Sysmon itself is free; data ingestion in Sentinel can lead to **significant costs** for all data. Be careful with the amount of data and used configuration.

Do we need Sysmon on all systems?

Sysmon deployment can be scoped in the environment. Ideally ingest additional events for the Tier-0 servers and business-critical servers first. When starting small; the scope can be increased easily and tracking ingestion cost/ performance is easily possible to fine-tune configurations to save cost and ingestion of events. Even when Sysmon is locally installed without ingestion in Sentinel there is still a huge benefit.

- SecurityAlert
- SecurityBaseline
- SecurityBaselineSummary
- SecurityDetection
- SecurityEvent
- WindowsFirewall
- SysmonEvent
- ProtectionStatus
- Update (when the Update Management solution isn't running)
- UpdateSummary (when the Update Management solution isn't running)

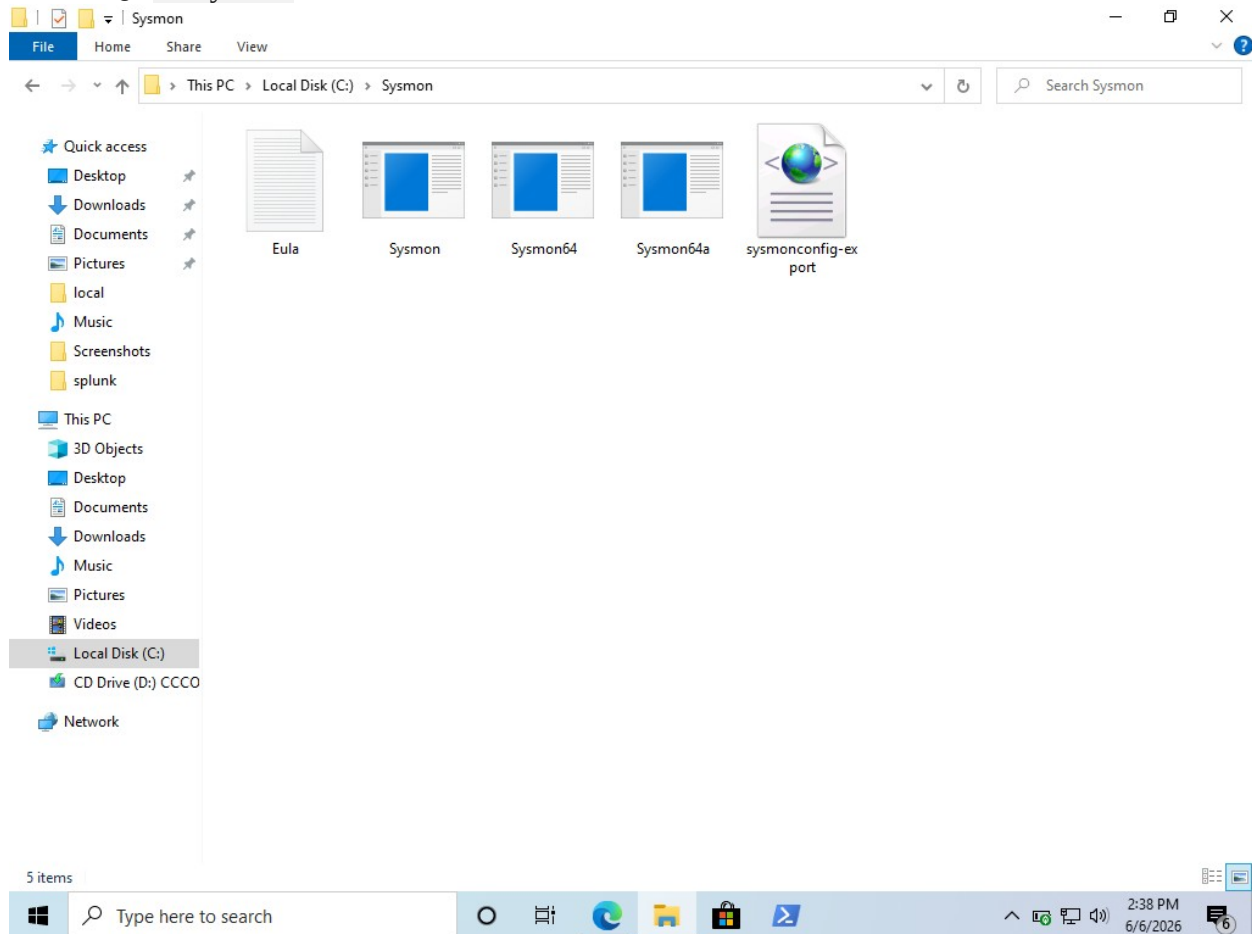
Installation and Setup Commands

Prerequisites

Download Sysmon: Download the official Sysmon tool from the [Microsoft Sysinternals Suite](#).

Download Configuration: Download the desired XML configuration file (e.g., sysmonconfig.xml from Olaf Hartong's repository).

Prepare Files: Extract the Sysmon ZIP file and place the Sysmon executable (Sysmon64.exe for 64-bit systems) and the downloaded configuration file (e.g., sysmon_config.xml) in the same easily accessible folder (e.g., C:\Sysmon).



Install Sysmon

System Monitor (Sysmon) is a Windows system service and device driver. For using Sysmon we need to have the following components:

- Sysmon installer
- Sysmon configuration file

Sysmon Installer

Sysmon can be downloaded directly from the [Microsoft site](#). Use the Sysmon(64).exe file for further deployment.

Configuration file

Sysmon needs to be installed in combination with the configuration file. Currently, there are multiple Sysmon configuration files available. Let's discuss the configuration file more in-depth in the next section.

Silent installation + configuration file

```
sysmon64.exe -accepteula -i sysmonconfiguration.xml
```

```
PS C:\Users\Administrator> cd C:\Users\Administrator\Downloads\Sysmon
PS C:\Users\Administrator\Downloads\Sysmon> dir

Directory: C:\Users\Administrator\Downloads\Sysmon

Mode                LastWriteTime         Length Name
----                -
-a----            11/6/2025   1:28 AM           7490 Eula.txt
-a----            11/6/2025   1:28 AM        8480560 Sysmon.exe
-a----            11/6/2025   1:28 AM        4563248 Sysmon64.exe
-a----            11/6/2025   1:28 AM        4993440 Sysmon64a.exe
-a----            11/6/2025   1:28 AM        253169 sysmonconfig.xml

PS C:\Users\Administrator\Downloads\Sysmon> .\Sysmon64.exe -i sysmonconfig.xml

System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.90
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
```

Installing Sysmon with the configuration file is a straightforward process.

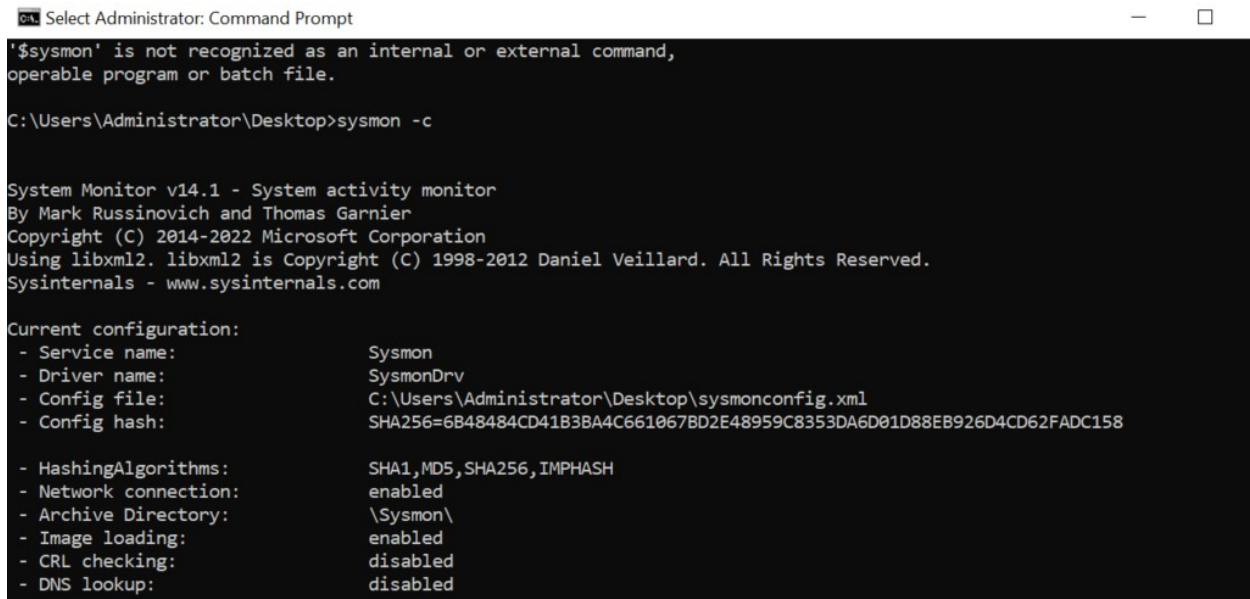
Validate Sysmon

```
##Show the configuration schema
```

```
sysmon -s
```

```
##Dump the current configuration
```

sysmon -c



```
Select Administrator: Command Prompt

'$sysmon' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator\Desktop>sysmon -c

System Monitor v14.1 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2022 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

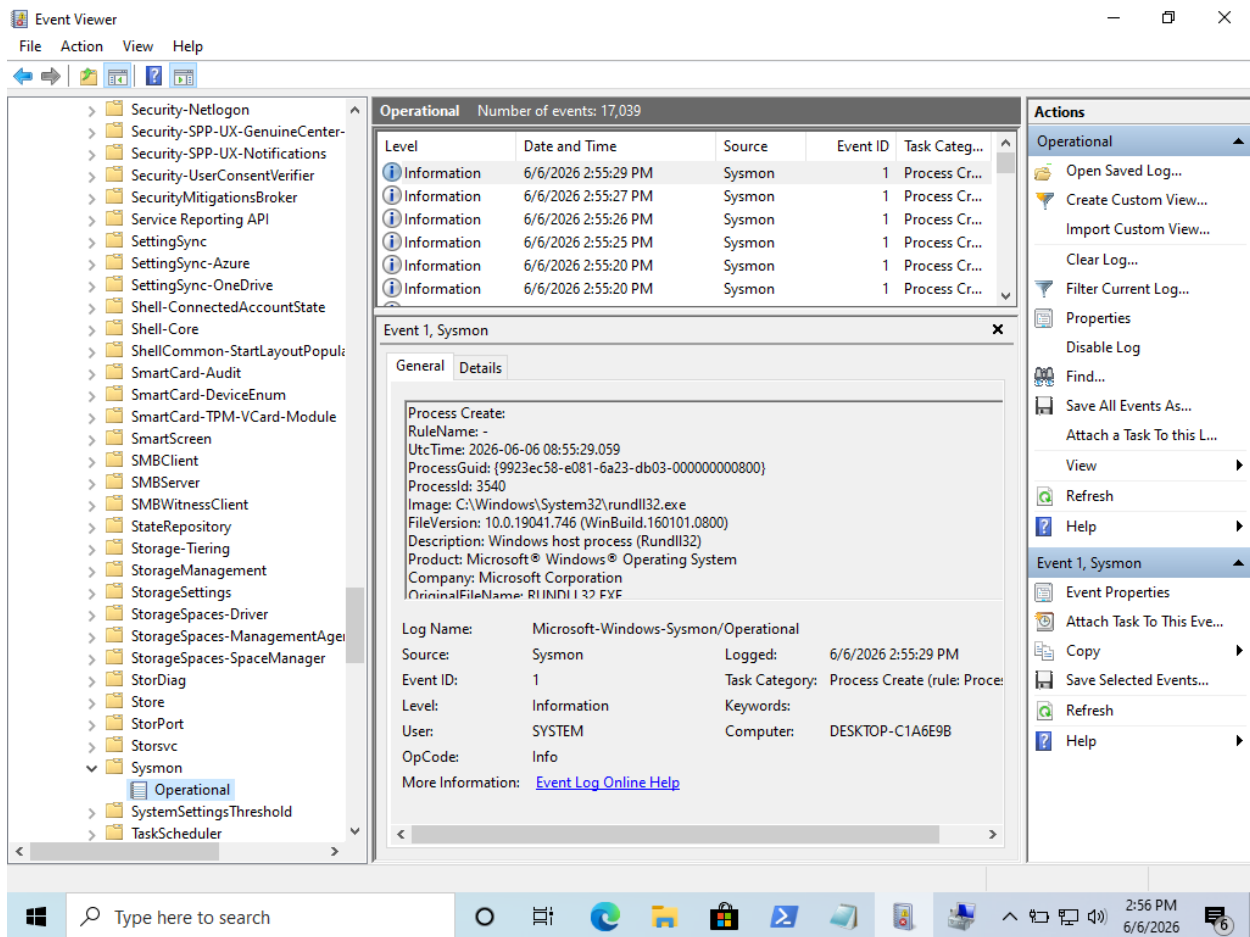
Current configuration:
- Service name: Sysmon
- Driver name: SysmonDrv
- Config file: C:\Users\Administrator\Desktop\sysmonconfig.xml
- Config hash: SHA256=6B48484CD41B3BA4C661067BD2E48959C8353DA6D01D88EB926D4CD62FADC158

- HashingAlgorithms: SHA1,MD5,SHA256,IMPHASH
- Network connection: enabled
- Archive Directory: \Sysmon\
- Image loading: enabled
- CRL checking: disabled
- DNS lookup: disabled
```

Verification

To confirm Sysmon is logging events:

- Open the **Event Viewer** (eventvwr.msc).
- Go to **Applications and Services Logs → Microsoft → Windows → Sysmon → Operational**
- You should immediately see **Event ID 1 (Process Creation)** events for the Sysmon service starting, and other events as you use the system.



PowerShell can be used for viewing the first 10 results

Viewing the first 10 Sysmon events in “Microsoft-Windows-Sysmon/Operational” via PowerShell

Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" | Select-Object -First 10

```
PS C:\Users\Administrator> Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" | Select-Object -First 10
>>

ProviderName: Microsoft-Windows-Sysmon

TimeCreated          Id LevelDisplayName Message
-----
10/8/2022 12:32:58 AM 13 Information Registry value set:...
10/8/2022 12:32:58 AM 7 Information Image loaded:...
10/8/2022 12:32:58 AM 7 Information Image loaded:...
10/8/2022 12:32:58 AM 17 Information Pipe Created:...
10/8/2022 12:32:58 AM 23 Information File Delete archived:...
10/8/2022 12:32:58 AM 7 Information Image loaded:...
10/8/2022 12:32:58 AM 11 Information File created:...
10/8/2022 12:32:58 AM 7 Information Image loaded:...
10/8/2022 12:32:58 AM 12 Information Registry object added or deleted:...
10/8/2022 12:32:58 AM 12 Information Registry object added or deleted:...
```

Sysmon is now silently working in the background, providing the rich telemetry needed for advanced threat hunting and incident responses



Atomic Red Team Step-by-Step Installation

What is Atomic Red Team?

[Atomic Red Team Official Project](#)

Atomic Red Team is an open-source security testing framework developed by [Red Canary](#). It allows organizations to safely simulate attacker behaviors based on the [MITRE ATT&CK Framework](#).

Instead of performing a full penetration test, Atomic Red Team executes small, controlled tests ("atomics") that mimic real-world adversary techniques.

Example

An attacker may use:

- PowerShell execution
- Credential dumping
- Scheduled tasks
- Registry modification
- Fileless malware techniques

Atomic Red Team provides predefined tests for these techniques so defenders can verify whether:

- Sysmon logs are generated
- Splunk detects the activity
- Wazuh triggers alerts
- SOC analysts receive notifications

Architecture

MITRE ATT&CK Technique

↓

Atomic Test

↓

Execution on Endpoint

↓

Sysmon Logs

↓

Splunk Universal Forwarder
↓
Splunk Enterprise
↓
Detection & Alert Validation

Why SOC Analysts Use Atomic Red Team

1. Threat Detection Validation

Tests whether SIEM rules detect known attack techniques.

2. Purple Teaming

Allows Red Team and Blue Team to collaborate.

3. SOC Training

Provides realistic attack simulations without deploying actual malware.

4. Detection Engineering

Helps create and tune:

- Splunk Correlation Searches
- Wazuh Rules
- Sigma Rules
- YARA Rules

5. Compliance & Audit Evidence

Demonstrates security monitoring effectiveness during audits.

Common Atomic Tests

MITRE Technique	Description
T1059	PowerShell Execution
T1003	Credential Dumping
T1547	Registry Persistence

MITRE Technique	Description
-----------------	-------------

T1053	Scheduled Task
T1110	Password Brute Force
T1021	Remote Services
T1070	Log Clearing
T1562	Defense Evasion

Atomic Red Team in Your Splunk Lab

For your planned lab:

Windows Victim



↓

Splunk Enterprise

↓

Detection Dashboard

Workflow

1. Install Sysmon
2. Install Splunk Universal Forwarder
3. Configure WinEventLog collection
4. Install Atomic Red Team
5. Execute Atomic Tests
6. Verify logs in Splunk
7. Build detections and dashboards

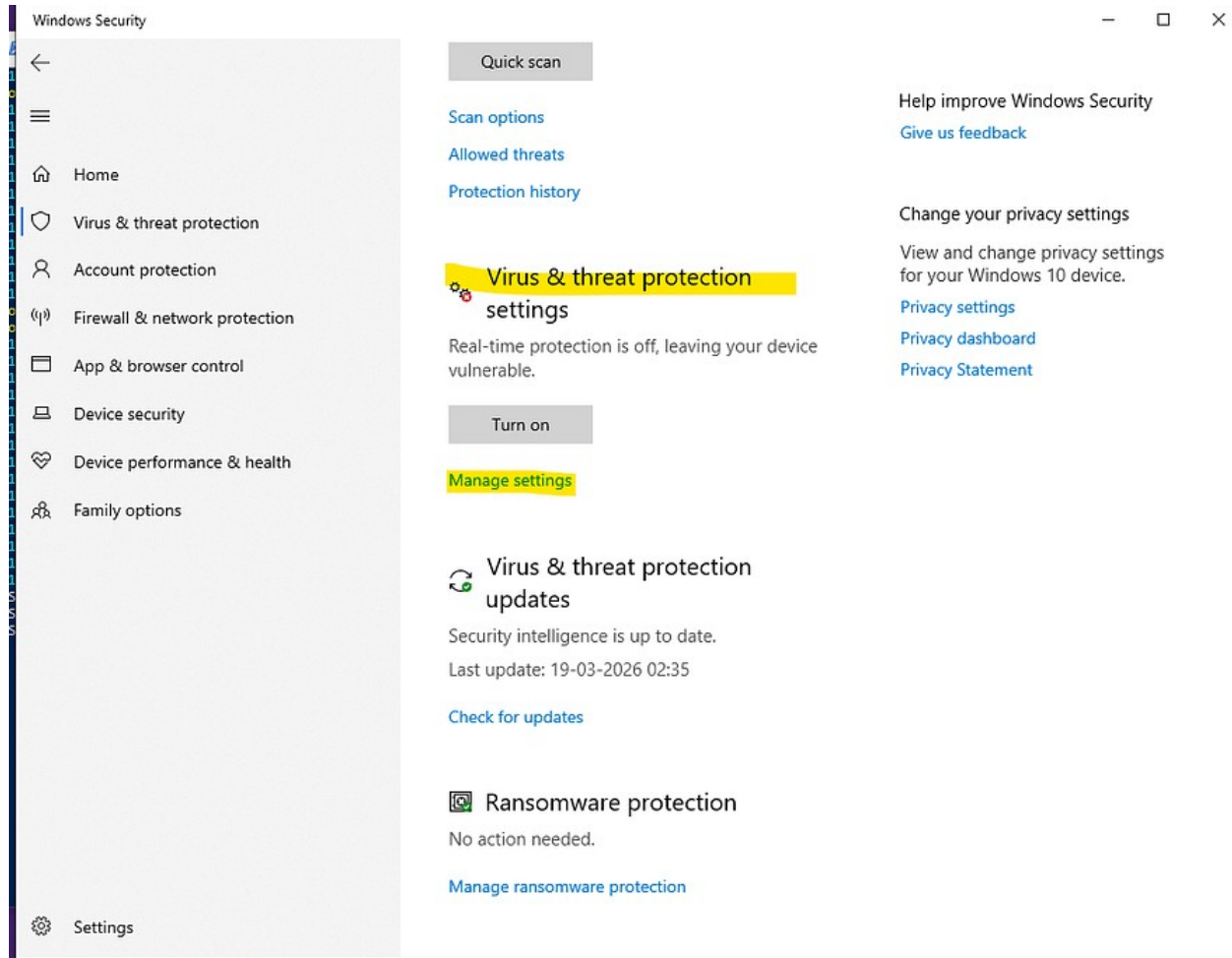
Installation

The installation process for tools or frameworks related to Atomic Red Team and MITRE ATT&CK can vary depending on the specific tools you are referring to. Here are general guidelines for the installation of two widely used components associated with these concepts:

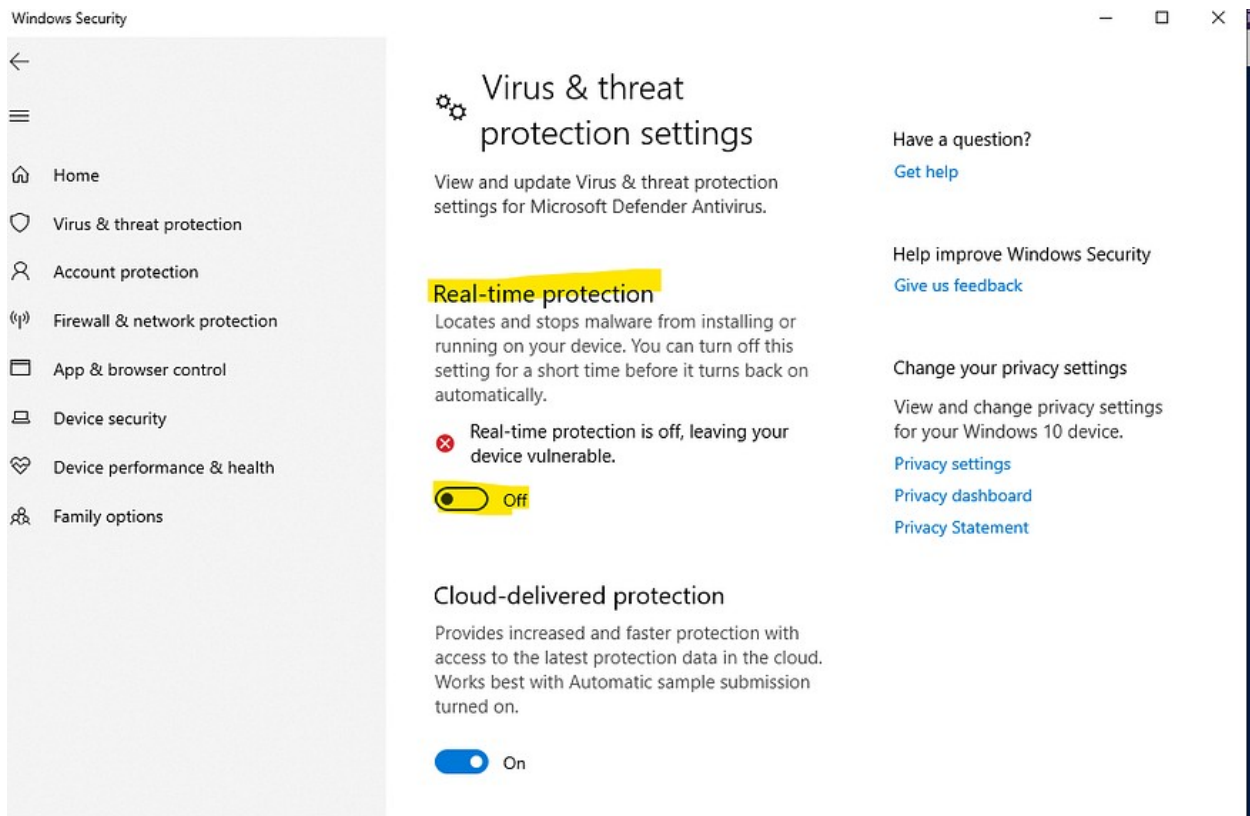
Step 1. Temporarily check Microsoft Defender settings

I opened Windows Security and went to Virus & threat protection settings

Why this matters: Defender can block or quarantine lab files while you are downloading or extracting them.

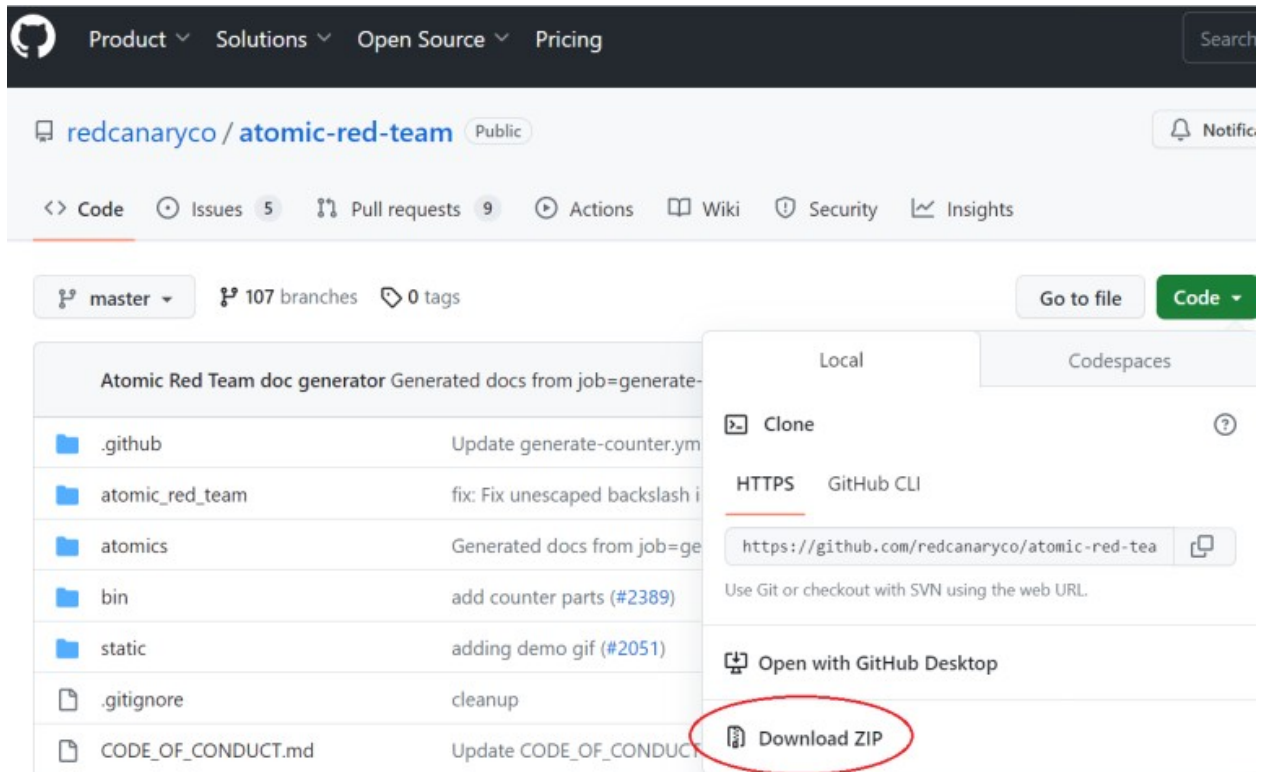


Open Virus & threat protection and then Manage settings



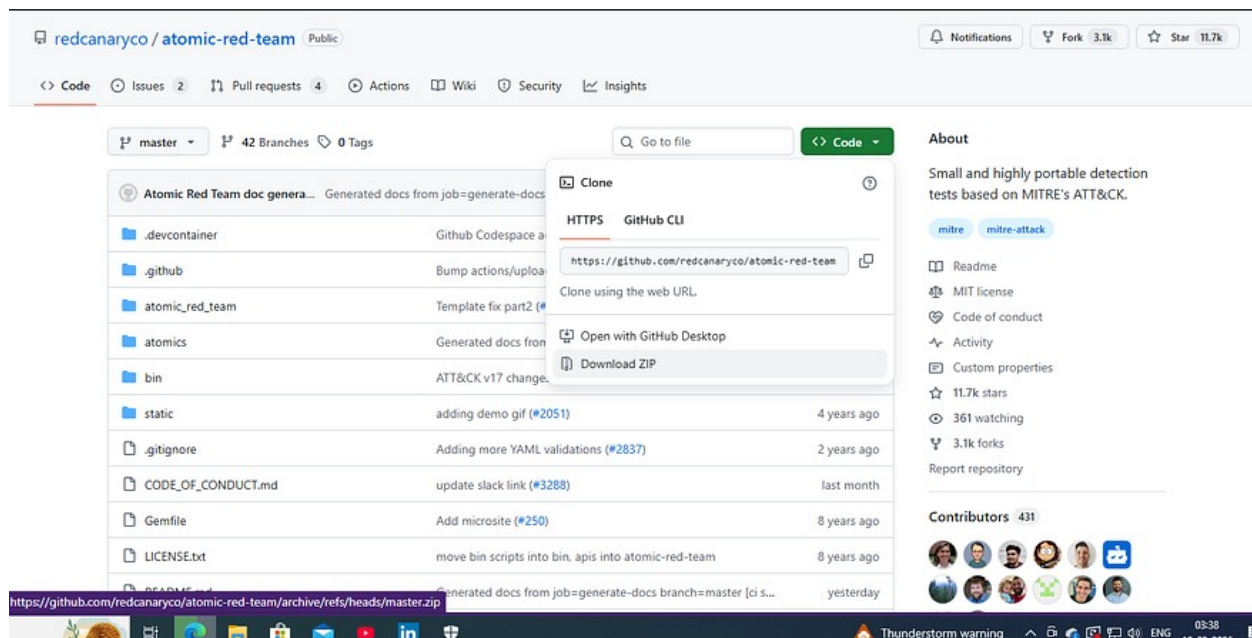
Check the Real-time protection status before continuing

Step 2. First, download Atomic Red Team from the link below
<https://github.com/redcanaryco/atomic-red-team>

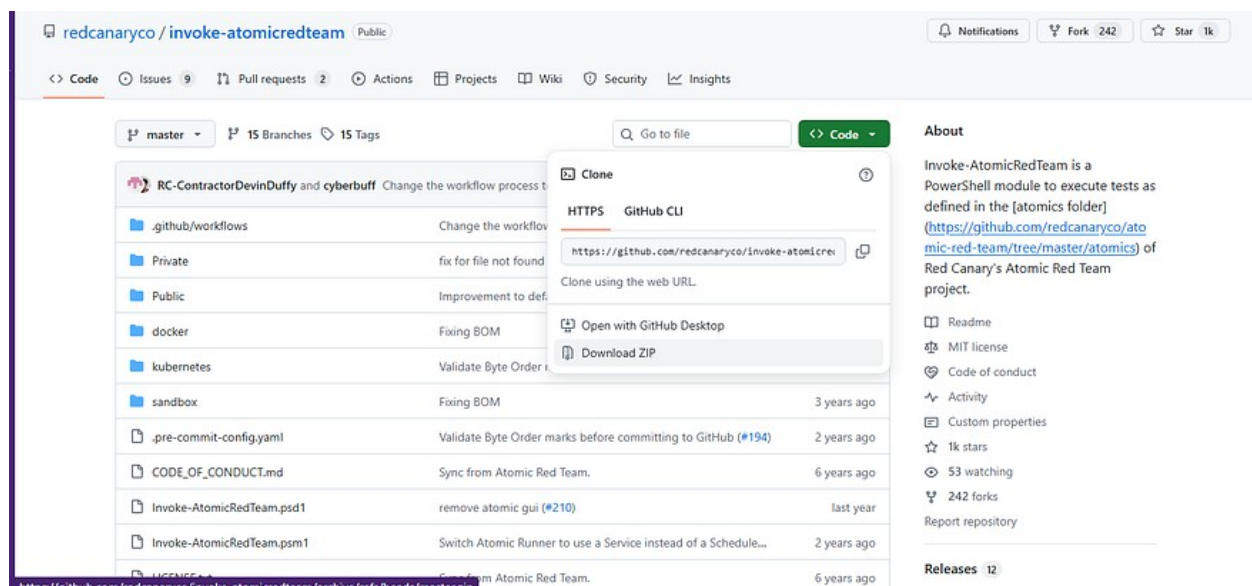


Step 3. Download files and create a local working folder

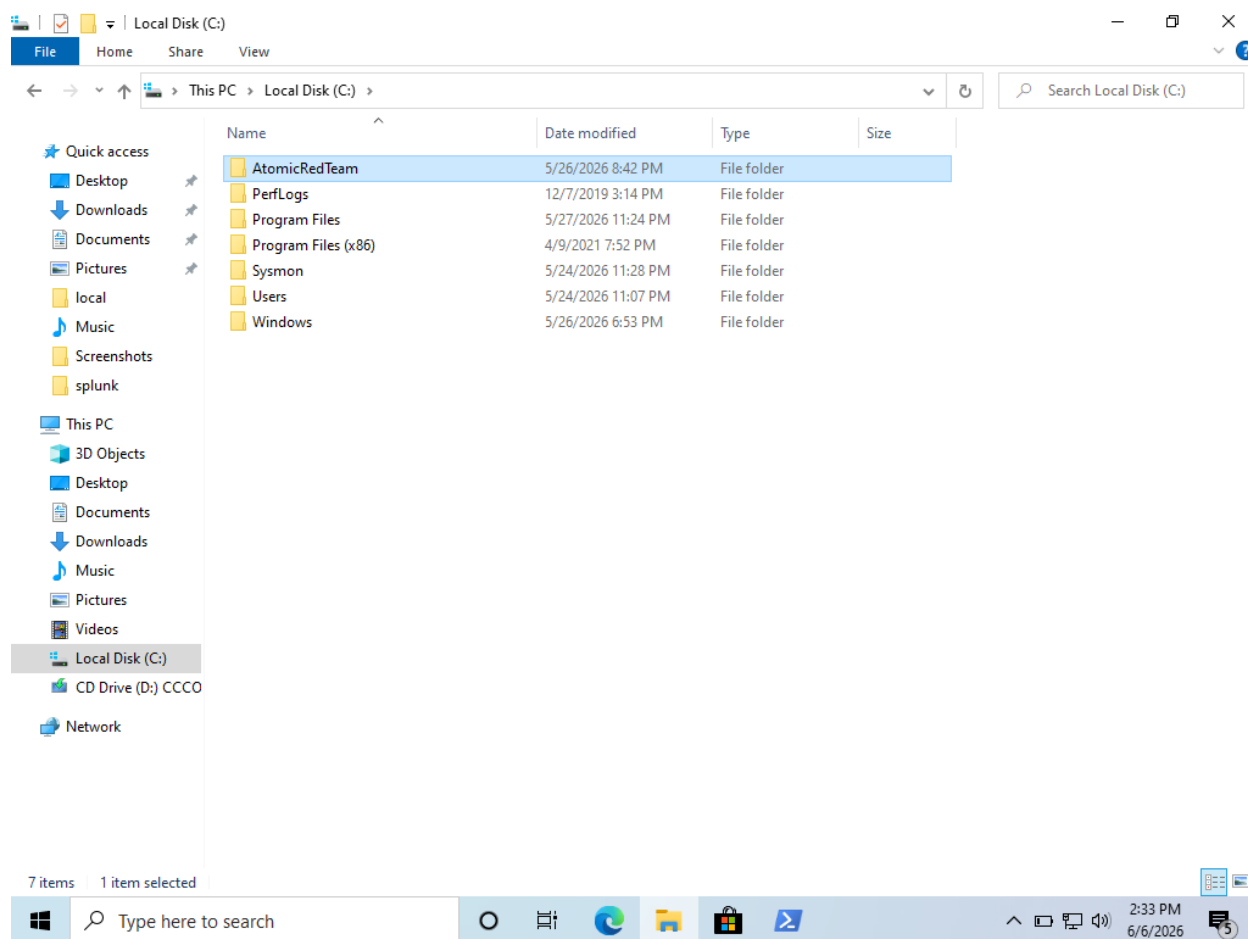
I downloaded [atomic-red-team](#) zip file and invoke-atomicredteam zip file repository and created a simple local folder on the C drive.(Note: keep folder name exactly as — AtomicRedTeam)



The atomic-red-team repository download page



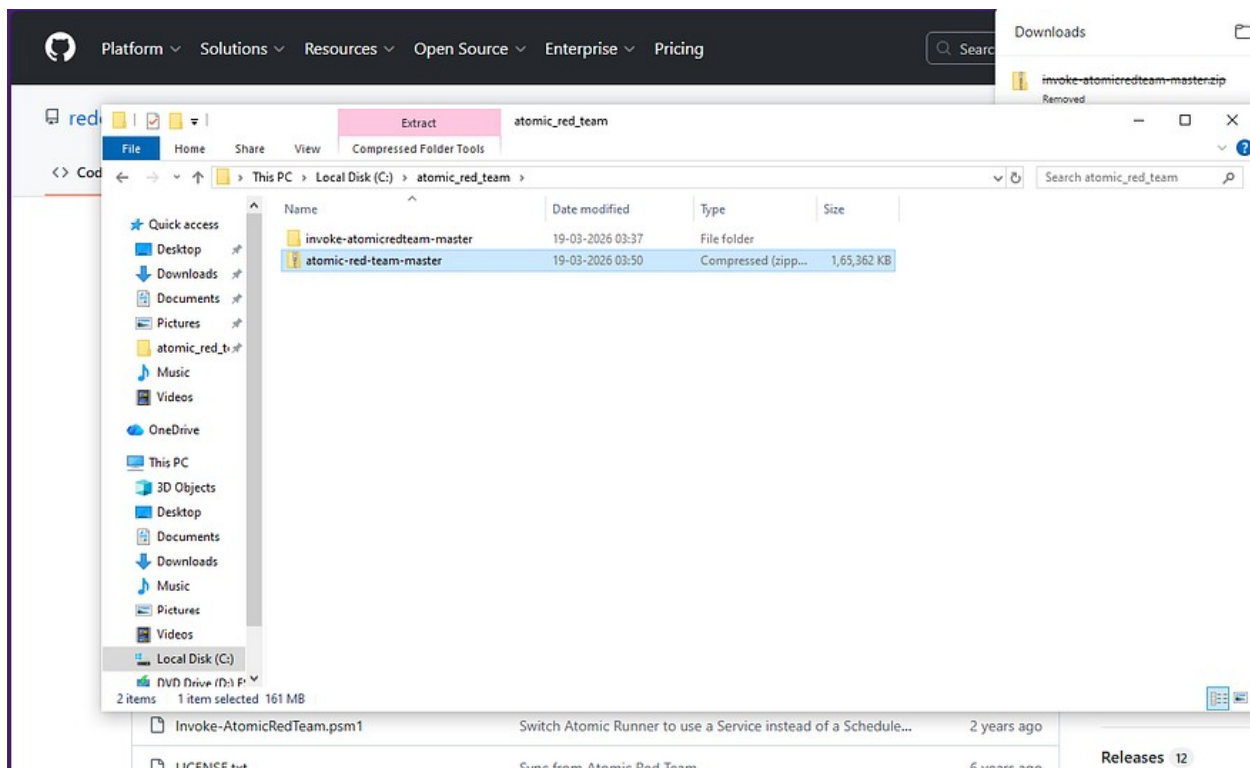
The invoke-atomicredteam repository



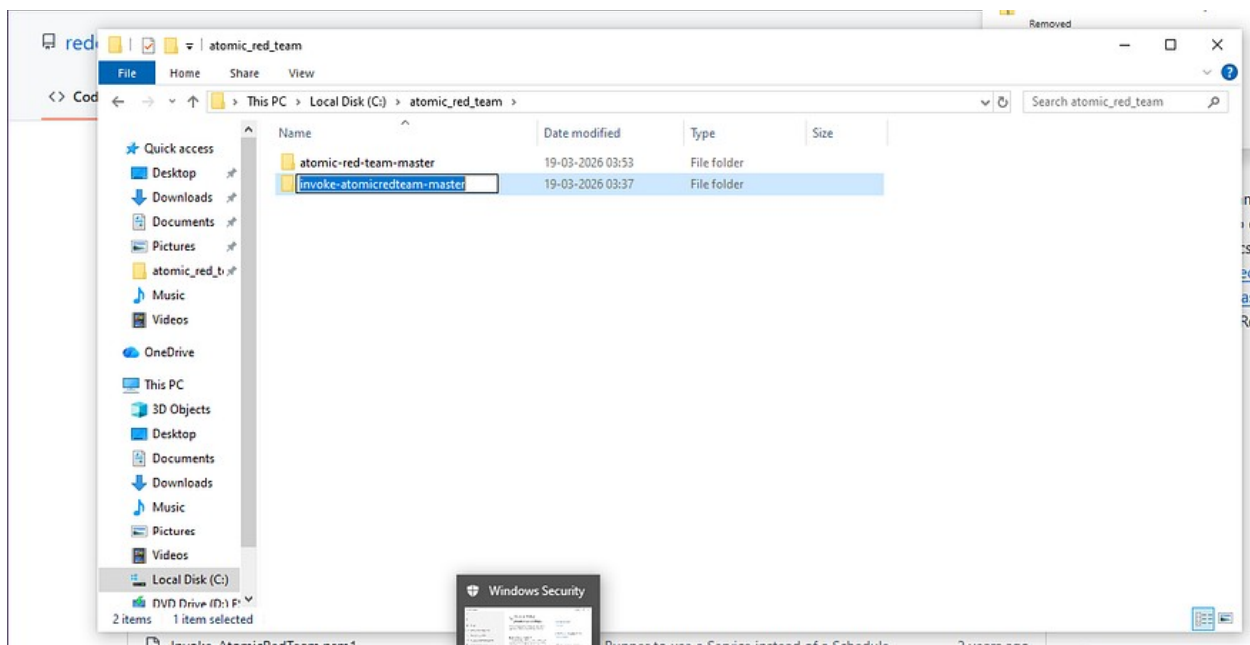
My local working folder on C:.

Step 4. Extract the downloaded files and identify the atomics folder

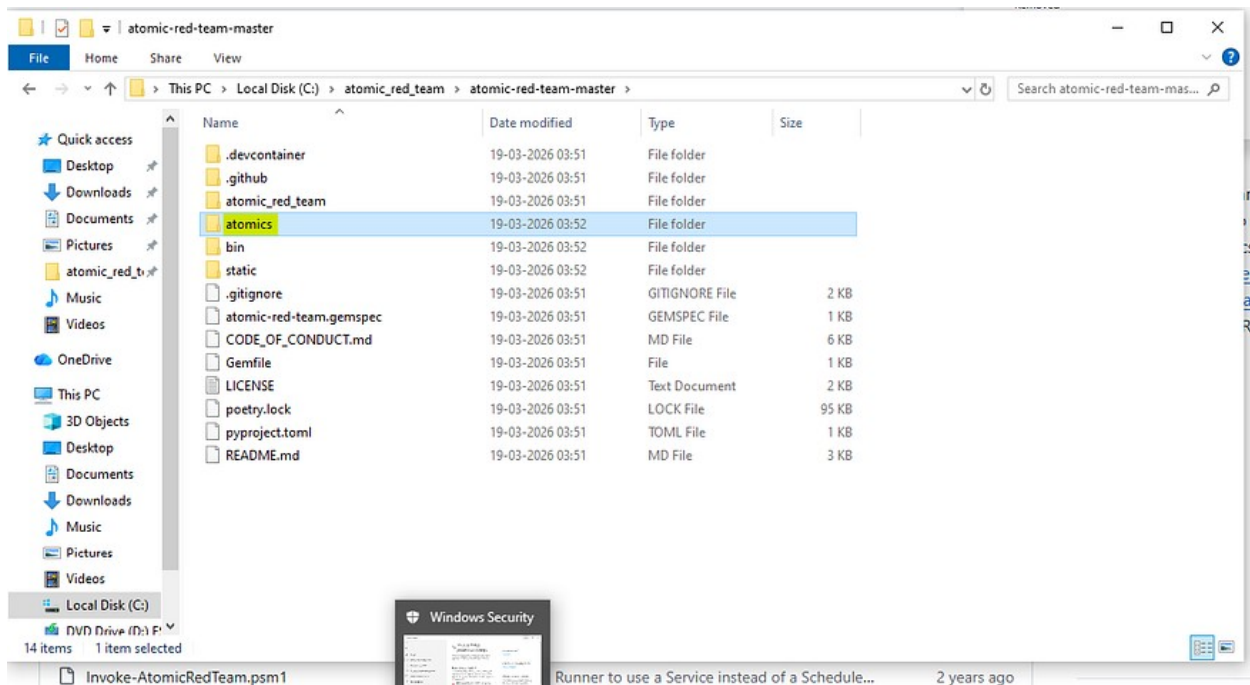
I extracted the downloaded ZIP and then opened the extracted content in a local working folder .The tests live inside the atomics folder, so I needed to locate it clearly before running commands.



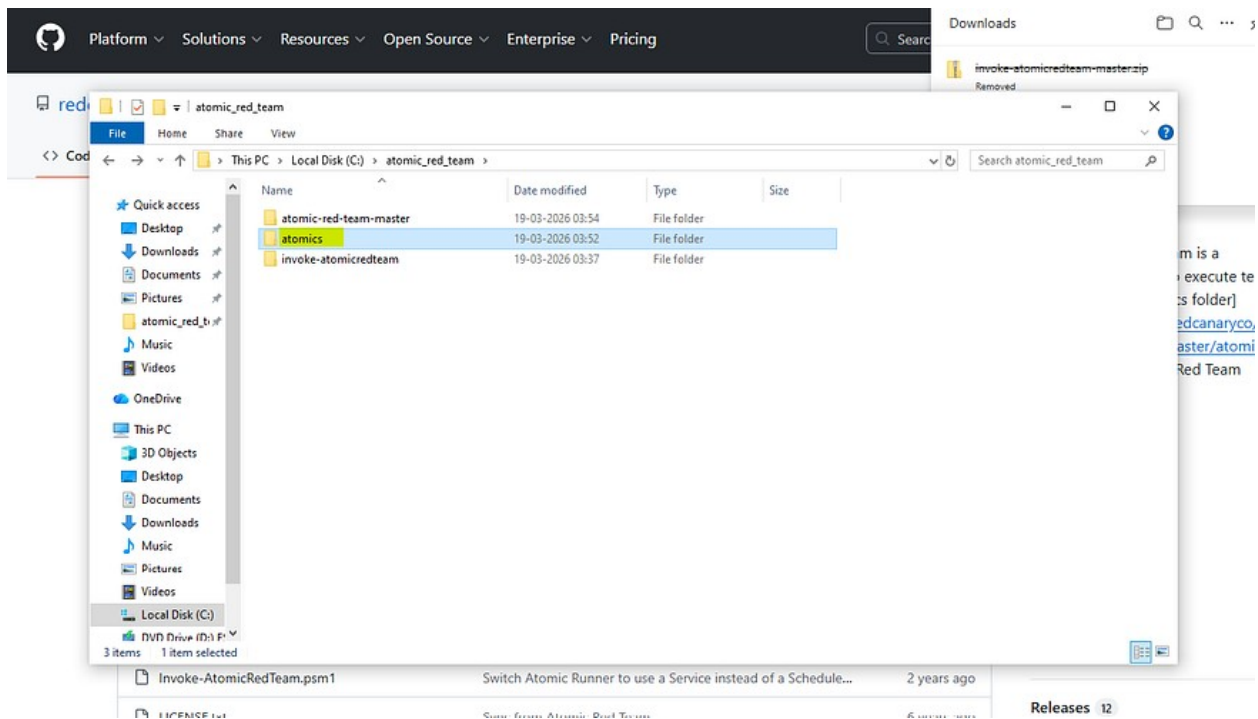
The downloaded ZIP inside my working folder.



After extraction, the folder becomes visible



Inside the extracted repository



The atomics folder that matters during execution

Step 6. Install the PowerShell module

Command used:

Install-Module -Name invoke-atomicredteam,powershell-yaml -Scope CurrentUser

I ran the install command in PowerShell. PowerShell may ask to install the NuGet provider and may also ask whether to trust PSGallery.

Install Execution Framework Only

To install the execution framework (Invoke-AtomicRedTeam) run the following command from a PowerShell prompt:

```
Invoke-Expression (IWR 'https://raw.githubusercontent.com/redcanaryco/invoke-atomicredteam/master/install-atomicredteam.ps1');  
Install-AtomicRedTeam
```

The documented install command

```
PS C:\Users\shail> Install-Module -Name invoke-atomicredteam,powershell-yaml -Scope CurrentUser  
  
NuGet provider is required to continue  
PowerShellGet requires NuGet provider version '2.8.5.201' or newer to interact with NuGet-based repositories. The NuGet  
provider must be available in 'C:\Program Files\PackageManagement\ProviderAssemblies' or  
'C:\Users\shail\AppData\Local\PackageManagement\ProviderAssemblies'. You can also install the NuGet provider by running  
'Install-PackageProvider -Name NuGet -MinimumVersion 2.8.5.201 -Force'. Do you want PowerShellGet to install and  
import the NuGet provider now?  
[Y] Yes [N] No [S] Suspend [?] Help (default is "Y"):
```

Get provider prompt during installation

```
Windows PowerShell
NuGet provider is required to continue

Installing 2 modules.
Installing Module 'Invoke-AtomicRedTeam' (1 of 2).
[0]
Installing package 'Invoke-AtomicRedTeam'
Installing dependent package 'powershell-yaml'
[ooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo]

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\shail> Install-Module -Name invoke-atomicredteam,powershell-yaml -Scope CurrentUser

NuGet provider is required to continue
PowerShellGet requires NuGet provider version '2.8.5.201' or newer to interact with NuGet-based repositories. The NuGet
provider must be available in 'C:\Program Files\PackageManagement\ProviderAssemblies' or
'C:\Users\shail\AppData\Local\PackageManagement\ProviderAssemblies'. You can also install the NuGet provider by running
'Install-PackageProvider -Name NuGet -MinimumVersion 2.8.5.201 -Force'. Do you want PowerShellGet to install and
import the NuGet provider now?
[Y] Yes [N] No [S] Suspend [?] Help (default is "Y"): y

Untrusted repository
You are installing the modules from an untrusted repository. If you trust this repository, change its InstallationPolicy value by
running the Set-PSRepository cmdlet. Are you sure you want to install the modules from 'PSGallery'?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): y
```

PS Gallery trust prompt and successful installation flow

Step 7. List tests and run a sample Atomic test

We can now launch attacks. by this command can test all of the atomic attacks on the server.

Invoke-AtomicTest All

```
PS C:\Users\Administrator> Invoke-AtomicTest All
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Highway to the danger zone, Executing All Atomic Tests!
Do you wish to execute all tests?
[Y] Yes [N] No [S] Suspend [?] Help (default is "Y"): y
```



```

PS C:\Users\Administrator> Invoke-AtomicTest t1003
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1003-1 Gsecdump
'C:\AtomicRedTeam\atomics\..\ExternalPayloads\gsecdump.exe' is not recognized as an internal or external command,
operable program or batch file.
Exit code: 1
Done executing test: T1003-1 Gsecdump
Executing test: T1003-2 Credential Dumping with NPPSpy
[!] Please, logout and log back in. Cleartext password for this account is going to be located in C:\NPPSpy.txt
Copy-Item : Cannot find path 'C:\AtomicRedTeam\ExternalPayloads\NPPSPY.dll' because it does not exist.
At line:1 char:4
+ & {Copy-Item "C:\AtomicRedTeam\atomics\..\ExternalPayloads\NPPSPY.dll ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\AtomicRedTea...oads\NPPSPY.dll:String) [Copy-Item], ItemNotFound
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.CopyItemCommand
Exit code: 0
Done executing test: T1003-2 Credential Dumping with NPPSpy
Executing test: T1003-3 Dump svchost.exe to gather RDP credentials
Exit code: 0
Done executing test: T1003-3 Dump svchost.exe to gather RDP credentials
Executing test: T1003-4 Retrieve Microsoft IIS Service Account Credentials Using AppCmd (using list)
C:\Windows\System32\inetsrv\appcmd.exe : The term 'C:\Windows\System32\inetsrv\appcmd.exe' is not recognized as th
name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was
included, verify that the path is correct and try again.
At line:1 char:4
+ & {C:\Windows\System32\inetsrv\appcmd.exe list apppool /@:*
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Windows\System32\inetsrv\appcmd.exe:String) [], CommandNotFoundE
+ FullyQualifiedErrorId : CommandNotFoundException

C:\Windows\System32\inetsrv\appcmd.exe : The term 'C:\Windows\System32\inetsrv\appcmd.exe' is not recognized as th
name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was
included, verify that the path is correct and try again.
At line:2 char:1
+ C:\Windows\System32\inetsrv\appcmd.exe list apppool /@text:*
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Windows\System32\inetsrv\appcmd.exe:String) [], CommandNotFoundE
+ FullyQualifiedErrorId : CommandNotFoundException

```

Invoke-AtomicTest T1003 –Show Details Brief

```

See Wiki at https://github.com/redcanaryco/Invoke-AtomicRedTeam/wiki for complete details
PS C:\Users\Administrator> Invoke-AtomicTest t1003 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

T1003-1 Gsecdump
T1003-2 Credential Dumping with NPPSpy
T1003-3 Dump svchost.exe to gather RDP credentials
T1003-4 Retrieve Microsoft IIS Service Account Credentials Using AppCmd (using list)
T1003-5 Retrieve Microsoft IIS Service Account Credentials Using AppCmd (using config)
T1003-6 Dump Credential Manager using keymgr.dll and rundll32.exe
PS C:\Users\Administrator> Invoke-AtomicTest t1003 -TestNumbers 1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1003-1 Gsecdump
The system cannot find the path specified.
Exit code: 1
Done executing test: T1003-1 Gsecdump
PS C:\Users\Administrator>

```

This combination is one of the best hands-on labs for learning:

- SOC Operations
- Threat Hunting

- Detection Engineering
- Incident Response
- MITRE ATT&CK Mapping
- IT Security Auditing

and aligns well with a career path toward Information Security, IT Risk Management, and IT Audit in the banking sector.

Therefore, by utilizing this tool, we can assess our preparedness for potential attacks across a range of tactics and techniques that our company or institution might encounter. This tool enables us to evaluate our security posture against these potential threats, providing an opportunity to identify and address security vulnerabilities before facing an actual cyber attack.